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**32384 CHARLES COUNTY POMONKEY PUMP STATION
IMPROVEMENTS**

INSTRUMENTATION SUBMITTAL

SECTION 17500 - INSTRUMENTATION

February 04, 2026

ENGINEER: WRA

CONTRACTOR: JOHNSTON CONSTRUCTION

***MANUFACTURERS' REPRESENTATIVE FOR ENGINEERED INSTRUMENTATION,
PROCESS EQUIPMENT AND PUMPING SYSTEMS***

Branch Offices: RICHMOND, VA and LEOLA, PA

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

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General Comments:

1. This submittal assumes cable lengths for the various instruments included.
Contractor to confirm cable lengths for instruments.

Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION
1	Endress + Hauser	FMR20-CBPBMVCEVEE3+Z1	Radar Level Transmitter
1	Endress + Hauser	71429910	Swiveling Mounting Bracket, Sewer
2	Anchor Scientific	GSE60NONC-GOLD	Float Switch, NO
3	Endress + Hauser	RSG35-B2A+C1	Digital Chart Recorder
3	Endress + Hauser	71187780	1GB Neutral SD Memorycard
4	Precision Digital	PD6000-7H2	Digital Process Meter Display

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Spare Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION	QTY
1	Endress + Hauser	FMR20- CBPBMVCEVEE3+Z1	Radar Level Transmitter	1
3	Endress + Hauser	71187780	1GB Neutral SD Memorycard	1

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Data Sheet 1

Customer:	Johnston Construction
Reference:	32384 Charles County Pomonkey Pump Station Improvements
Date:	02/04/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Endress + Hauser: Radar Level Transmitter Model Number: FMR20-CBPBMVCEVEE3+Z1 Tags/Service: LT-01 / Pump Station Wet Well Specifications: • Up to 66ft measuring range • FMR20 - Micropilot FMR20 series time-of-flight radar measurement • CB - CSA C/US IS CL.I Div.2 Gr.A-d, AEx/Ex ia IIC T4 approvals • P - 2 wire power supply, 4-20mA HART output, HART/Bluetooth (App) configuration operation • BM - 40mm/1-$\frac{1}{2}$" antenna, 15m liquid, -40°C...80°C max measuring range • VCE - Thread ASME MNPT1 process connection rear side, PVDF material, FNPT $\frac{1}{2}$ conduit connection • VEE - Thread ASME MNPT1-$\frac{1}{2}$ process connection front side, PVDF material • 3 - 20m/65ft cable length • Z1 - Tagging
1	Mfg: Endress + Hauser: Swiveling Mounting Bracket, Sewer Model Number: 71429910

	Tags/Service: LT-01 / Pump Station Wet Well Specifications: • Pivotal and adjustable arm for mounting E+H FMR20 • Allows alignment of transmitter with the center of a channel
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RADAR LEVEL TRANSMITTER

TAGS: LT-01

Technical Information Micropilot FMR20 HART

Free space radar

Level measurement for liquids



Application

- Ingress protection: IP66/68 / NEMA 4X/6P
- Measuring range: up to 20 m (66 ft)
- Process temperature: -40 to 80 °C (-40 to 176 °F)
- Process pressure: -1 to 3 bar (-14 to 43 psi)
- Accuracy: up to ± 2 mm (0.08 in)
- International explosion protection certificates

Your benefits

- Level measurement for liquids in storage tanks, open basins, pump shafts and canal systems
- Radar measuring device with *Bluetooth*® wireless technology and HART communication
- Simple, safe and secure wireless remote access – ideal for installation in hazardous areas or places difficult to reach
- Commissioning, operation and maintenance via free iOS / Android app SmartBlue – saves time and reduces costs
- Full PVDF body - for a long sensor lifetime
- Hermetically sealed wiring and fully potted electronics – eliminates water ingress and allows operation under harsh environmental conditions
- Most compact radar due to unique radar chip design – fits in limited space installations
- Best price-performance-ratio radar

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Important document information

Symbols used

Symbols for certain types of information and graphics

-  **Permitted**
Procedures, processes or actions that are permitted
-  **Forbidden**
Procedures, processes or actions that are forbidden
-  **Tip**
Indicates additional information
-  **Reference to documentation**
-  **Reference to graphic**
-  **Notice or individual step to be observed**
-  **Series of steps**
-  **Result of a step**
-  **Item numbers**
-  **Views**

Terms and abbreviations

BA

Document type "Operating Instructions"

KA

Document type "Brief Operating Instructions"

TI

Document type "Technical Information"

SD

Document type "Special Documentation"

XA

Document type "Safety Instructions"

PN

Nominal pressure

MWP

MWP (Maximum working pressure/max. process pressure)

The MWP can also be found on the nameplate.

ToF

Time of Flight

FieldCare

Scalable software tool for device configuration and integrated plant asset management solutions

DeviceCare

Universal configuration software for Endress+Hauser HART, PROFIBUS, FOUNDATION Fieldbus and Ethernet field devices

DTM

Device Type Manager

 ϵ_r (Dk value)

Relative dielectric constant

Operating tool

The term "operating tool" is used in place of the following operating software:

- FieldCare / DeviceCare, for operation via HART communication and PC
- SmartBlue (app), for operation using an Android or iOS smartphone or tablet

BD

Blocking Distance; no signals are analyzed within the BD.

PLC

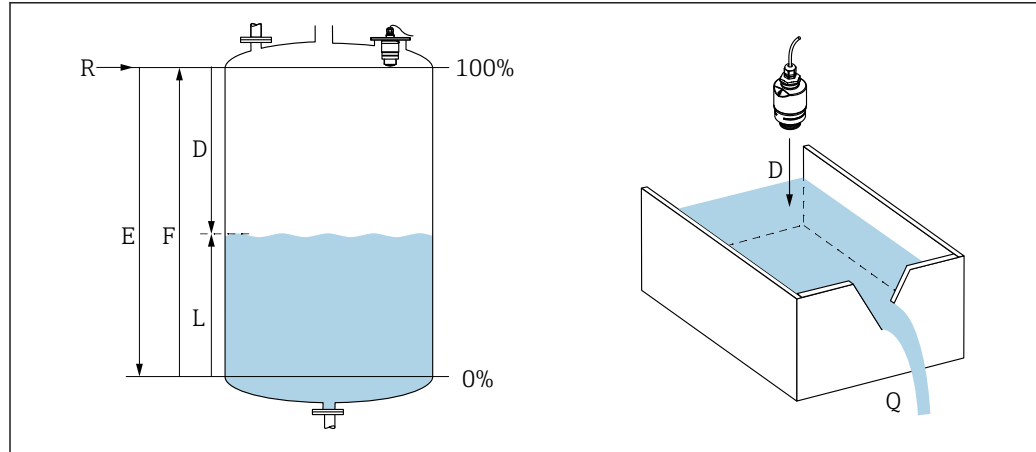
Programmable logic controller (PLC)

Product life cycle

Engineering	■ Proven radar measuring technology ■ Level measurement and open channel flow measurement for Ex and non-Ex areas ■ Flooding detection ■ Wide range of installation possibilities and accessories ■ Highest degree of ingress protection ■ 2D/3D drawings ■ Spec Sheet Producer ■ Applicator Selection tool for selecting the perfect measurement solution  Device not compatible with transmitters and sensors that use ultrasonic measurement technology (e.g. Prosonic FMU9x, FDU9x)
Procurement	■ Best price-performance-ratio radar ■ Global availability ■ Order code includes variety of mounting accessories and decentralized RIA15 process indicator for HART
Installation	■ Rear and front thread for flexible installation ■ Slip-on flange for nozzle installation ■ Complete measuring point: including mounting accessory, RIA15 and flooding protection tube
Commissioning	■ Quick and easy setup with the SmartBlue app and DeviceCare / FieldCare or RIA15 ■ No additional tools or adapters required ■ Local languages (up to 15)
Operation	■ Continuous self-monitoring ■ Diagnostics information according to NAMUR NE107 with remedial measures in the form of plain text messages ■ Signal curve via SmartBlue (app) and DeviceCare / FieldCare ■ Encrypted single point-to-point data transmission (tested by Fraunhofer Institute) and password-protected communication via <i>Bluetooth®</i> wireless technology
Maintenance	■ No maintenance required ■ Technical experts on-call around the world
Retirement	■ Environmentally responsible recycling concepts ■ RoHS compliance (restriction of certain hazardous substances), lead-free soldering of electronic components

Measuring principle

The Micropilot is a "downward-looking" measuring system, which functions according to the time-of-flight (ToF) method. It measures the distance from the reference point **R** to the product surface. Radar pulses are emitted by an antenna, reflected off the product surface and received again by the radar system.



A0028409

1 Level / flow calibration parameter

E Empty calibration (= zero)

F Full calibration (= span)

D Measured distance

L Level ($L = E - D$)

Q Flow rate at measuring weirs or channels (calculated from the level using linearization)

R Reference point

Input

The reflected radar pulses are received by the antenna and transmitted to the electronics. A microprocessor evaluates the signals and identifies the level echo caused by the reflection of the radar pulses at the product surface. This clear signal detection system benefits from over 30 years' experience with time-of-flight procedures.

The distance **D** to the product surface is proportional to the time of flight **t** of the pulse:

$$D = c \cdot t / 2,$$

where **c** is the speed of light.

Based on the known empty distance **E**, the level **L** is calculated:

$$L = E - D$$

Output

The Micropilot is calibrated by entering the empty distance **E** (= zero point) and the full distance **F** (= span).

- Current output: 4 to 20 mA
- Digital output (HART, SmartBlue): 0 to 15 m (0 to 49 ft)¹⁾ or 0 to 20 m (0 to 66 ft) depending on antenna version

1) order code 010 (approval) "GA", "GB", "GR": 0 to 10 m (0 to 32.8 ft)

Input

Measured variable	The measured variable is the distance between the reference point and the product surface. The level is calculated based on E , the empty distance entered.
Measuring range	Maximum measuring range → - Device with 40 mm (1.5 in) antenna: 15 m (49 ft) ²⁾ Device with 80 mm (3 in) antenna: 20 m (66 ft) Installation requirements - Recommended tank height > 1.5 m (5 ft) for media with low ϵ_r value - Open channel minimum width 0.5 m (1.6 ft) - Calm surfaces - No agitators - No buildup - Relative dielectric constant $\epsilon_r > 4$ Contact Endress+Hauser for lower ϵ_r values Usable measuring range The usable measuring range depends on the antenna size, the medium's reflective properties, the installation position and any possible interference reflections. Media groups $\epsilon_r = 4$ to 10 e.g. concentrated acid, organic solvents, ester, aniline, alcohol, acetone. $\epsilon_r > 10$ e.g. conductive liquids, aqueous solutions, diluted acids and bases Reduction of the maximum possible measuring range by: - Media with bad reflective properties (= low ϵ_r value) - Formation of buildup, particularly of moist products - Strong condensation - Foam generation - Freezing of sensor
Operating frequency	K-band (~ 26 GHz)
Transmission power	Mean power density in the direction of the beam - At a distance of 1 m (3.3 ft): < 12 nW/cm² - At a distance of 5 m (16 ft): < 0.4 nW/cm²

Output

Output signal	4 to 20 mA An 4 to 20 mA interface is used for measured value output and to power to the device.
Digital output	HART® - Signal encoding; FSK ± 0.5 mA over current signal - Data transmission rate; 1 200 Bit/s Bluetooth® wireless technology (available as an optional extra) The device has a <i>Bluetooth®</i> wireless technology interface and can be operated and configured via this interface using the SmartBlue app.

2) order code 010 (approval) "GA", "GB", "GR": 10 m (32.8 ft)

- The range under reference conditions is 25 m (82 ft)
- Incorrect operation by unauthorized persons is prevented by means of encrypted communication and password encryption
- The *Bluetooth*® wireless technology interface can be deactivated

Signal on alarm

Depending on the interface, failure information is displayed as follows:

- Current output
Alarm current: 22.5 mA (in accordance with NAMUR recommendation NE 43)
- Operating tool via digital communication (HART) or SmartBlue (app)
 - Status signal (as per NAMUR Recommendation NE 107)
 - Plain text display with remedial action

Linearization

The linearization function of the device allows the conversion of the measured value into any unit of length, weight, flow or volume. In DeviceCare and FieldCare, there are preprogrammed linearization tables for volume calculation in vessels.

Pre-programmed linearization curves

- Horizontal cylindrical tank
- Sphere
- Tank with pyramid bottom
- Tank with conical bottom
- Tank with flat bottom

Other linearization tables of up to 32 value pairs can be entered manually.

Protocol-specific data, HART

Manufacturer ID

17 (0x11)

Device type ID

44 (0x112c)

HART specification

7.0

Device description files (DTM)

Information and files under:

- www.endress.com
- www.hartcomm.org

HART load

Min. 250 Ω

HART device variables

Assignment of HART device variables is fixed and cannot be changed.

- **Measured values for PV (primary variable)**
Level linearized
- **Advanced diag. measured values for SV (secondary variable)**
Distance
- **Advanced diag. measured values for TV (tertiary variable)**
Relative echo amplitude
- **Advanced diag. measured values for QV (quarternary variable)**
Temperature

Supported functions

Additional transmitter status

Multidrop current

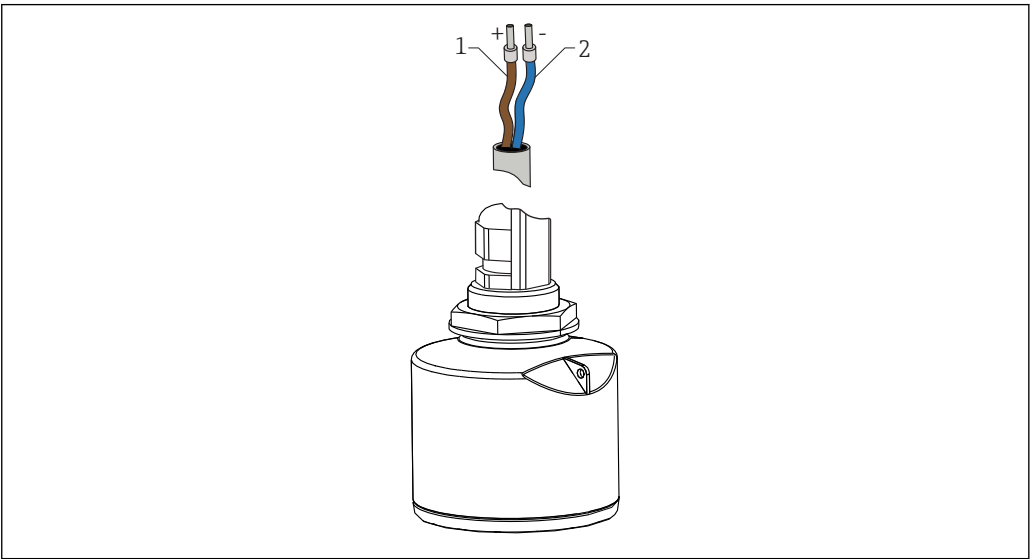
4 mA

Time for connection setup

< 1 s

Electrical connection

Cable assignment



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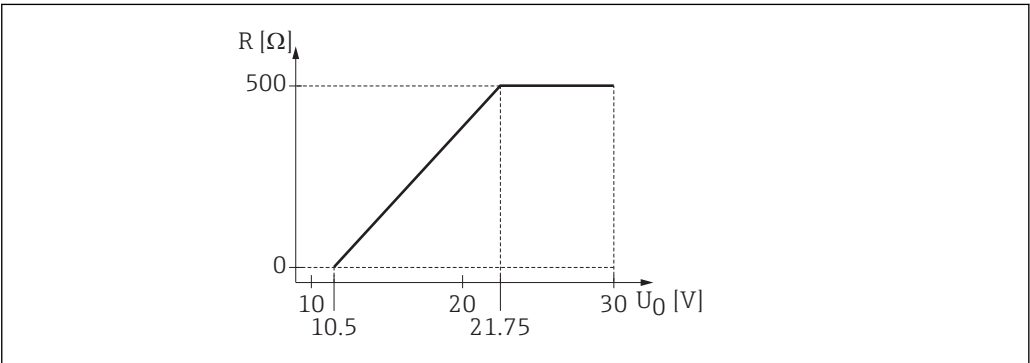
2 Cable assignment

- 1 Plus, brown wire
- 2 Minus, blue wire

Supply voltage

10.5 to 30 V_{DC}

An external power supply is necessary.



A0029226

3 Maximum load R , depending on supply voltage U_0 of power supply unit

Battery operation

The sensor's *Bluetooth*® wireless technology communication can be disabled to increase the operating life of the battery.

Potential equalization

No special measures for potential equalization are required.



Various power supply units can be ordered as an accessory from Endress+Hauser.

Power consumption

Maximum input power: 675 mW

Current consumption

- Maximum input current: <25 mA
- Maximum start-up current: 3.6 mA

Starting time

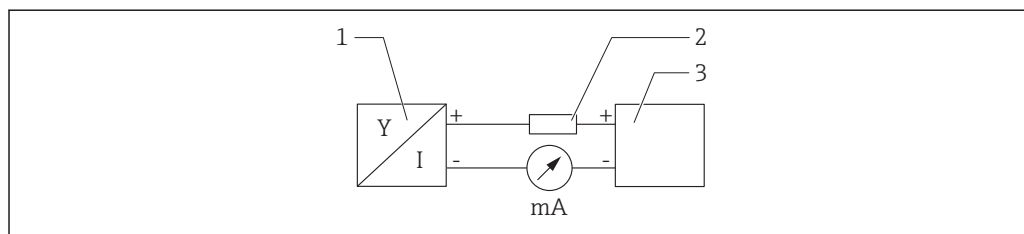
First stable measured value after 20 s (with supply voltage = 24 V_{DC})

Power supply failure

The configuration remains stored in the sensor.

Connecting the device**4 to 20 mA HART block diagram**

Connection of the device with HART communication, power source and 4 to 20 mA display



A0028908

 4 Block diagram of HART connection

1 Device with HART communication

2 HART resistor

3 Power supply



The HART communication resistor of $250\ \Omega$ in the signal line is always necessary in the case of a low-impedance power supply.

The voltage drop to be taken into account is:

Max. 6 V for $250\ \Omega$ communication resistor

HART device block diagram, connection with RIA15

FMR20 with RIA15 (incl. option for FMR20 basic configuration)



The RIA15 remote indicator can be ordered together with the device.

Product structure, feature 620 "Accessory enclosed":

- Option R4 "Remote indicator RIA15 non-hazardous area, field housing"
- Option R5 "Remote indicator RIA15 with explosion protection approval, field housing"



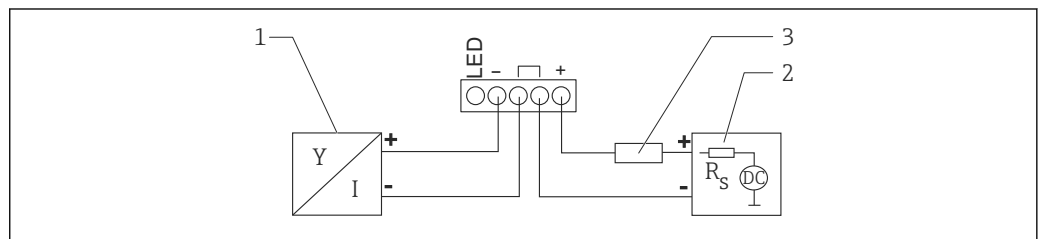
Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

Terminal assignment RIA15+
Positive connection, current measurement-
Negative connection, current measurement (without backlighting)LED
Negative connection, current measurement (with backlighting) $\perp$
Functional grounding: Terminal in housing

The RIA15 process indicator is loop-powered and does not require any external power supply.

The voltage drop to be taken into account is:

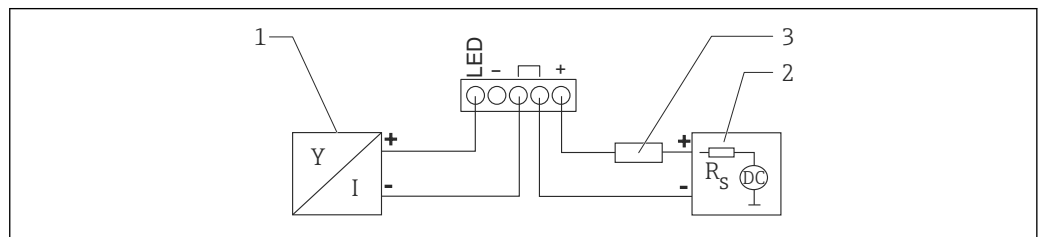
- ≤ 1 V in the standard version with 4 to 20 mA communication
- ≤ 1.9 V with HART communication
- and an additional 2.9 V if display light is used

Connection of the HART device and RIA15 without backlighting

A0019567

5 Block diagram of HART device with RIA15 process indicator without light

- 1 Device with HART communication
- 2 Power supply
- 3 HART resistor

Connection of the HART device and RIA15 with backlighting

A0019568

6 Block diagram of HART device with RIA15 process indicator with light

- 1 Device with HART communication
- 2 Power supply
- 3 HART resistor

Block diagram of HART device, RIA15 with installed HART communication resistor module

 The HART communication module for installation in the RIA15 can be ordered together with the device.

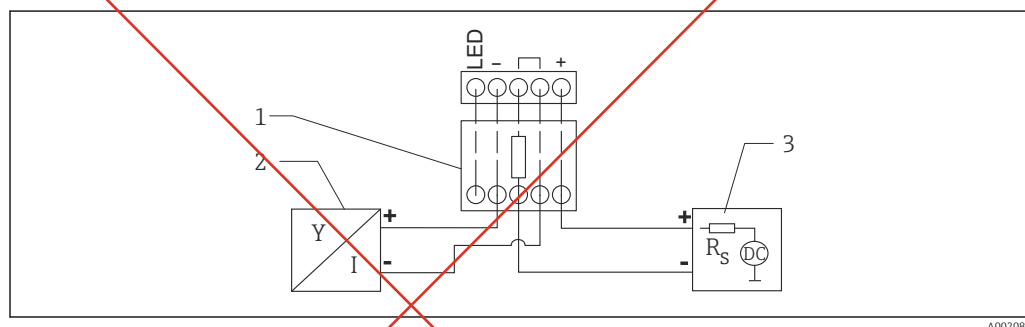
Product structure, feature 620 "Accessory enclosed":

Option R6 "HART communication resistor hazardous / non-hazardous area"

The voltage drop to be taken into account is:

Max. 7 V

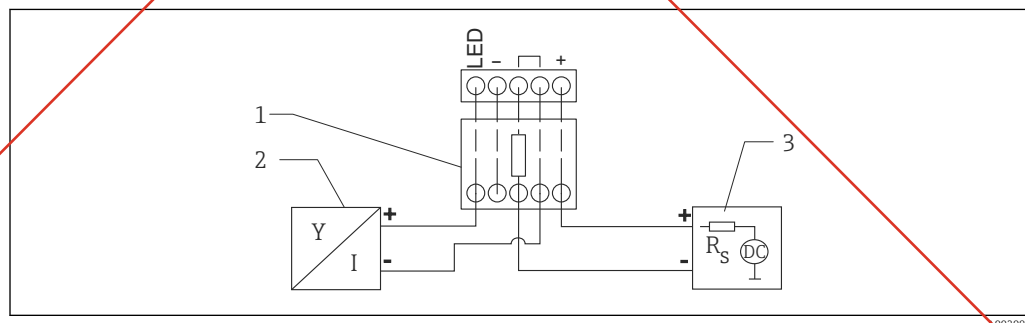
 Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

Connection of the HART communication resistor module, RIA15 without backlighting

A0020839

 7 Block diagram of HART device, RIA15 without light, HART communication resistor module

- 1 HART communication resistor module
- 2 Device with HART communication
- 3 Power supply

Connection of the HART communication resistor module, RIA15 with backlighting

A0020840

 8 Block diagram of HART device, RIA15 with light, HART communication resistor module

- 1 HART communication resistor module
- 2 Device with HART communication
- 3 Power supply

Cable specification

Unshielded cable, wire cross-section 0.75 mm²

- Resistant to UV and weather conditions as per ISO 4892-2
- Flame resistance according to IEC 60332-1-2

As per IEC/EN 60079-11 section 10.9, the cable is designed for a tensile strength of 30 N (6.74 lbf) (over a period of 1 h).

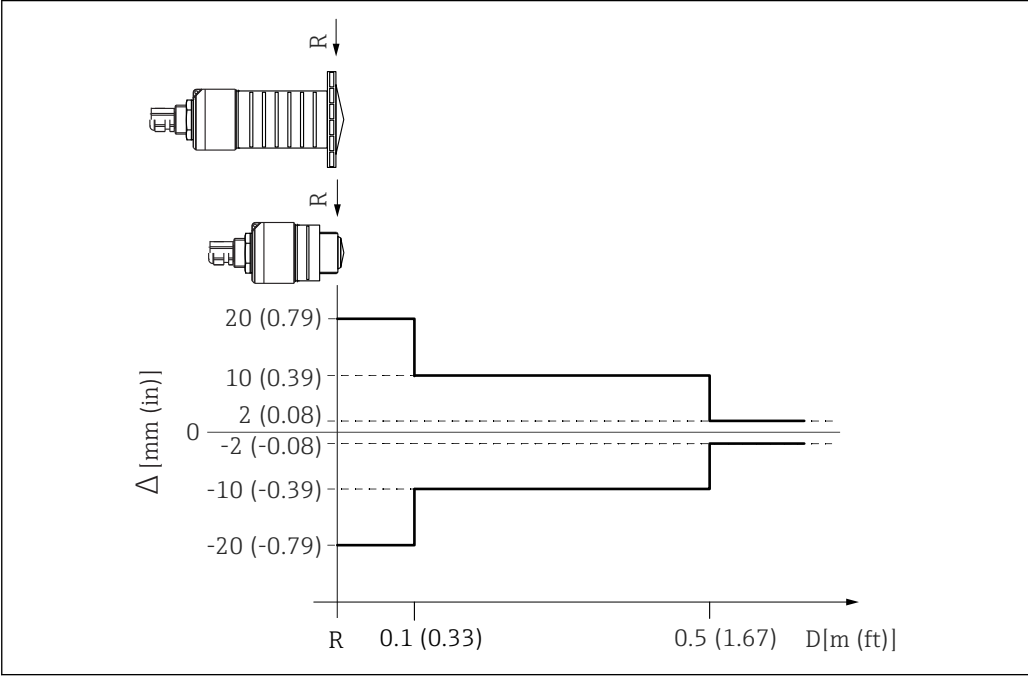
The device is supplied with 5 m (16 ft) cable length as standard. Cable lengths 10 m (33 ft) and 20 m (66 ft) are optionally available.

Lengths can be selected by the user up to an overall length of 300 m (980 ft) and are available by the meter (order option "8") or foot (order option "A").

Overvoltage protection

The device is equipped with integrated overvoltage protection.

Performance characteristics

Reference operating conditions	- Temperature = +24 °C (+75 °F) ±5 °C (±9 °F) - Pressure = 960 mbar abs. (14 psia) ±100 mbar (±1.45 psi) - Humidity = 60 % ±15 % - Reflector: metal plate with diameter ≥ 1 m (40 in) - No major interference reflections inside the signal beam
Maximum measured error	Typical data under reference operating conditions: DIN EN 61298-2, percentage values in relation to the span. Output, digital (HART, SmartBlue (app)) - Sum of non-linearity, non-repeatability and hysteresis: ±2 mm (±0.08 in) - Offset/zero point: ±4 mm (±0.16 in) Output, analog Only relevant for 4-20mA current output; add error of the analog value to the digital value - Sum of non-linearity, non-repeatability and hysteresis: ±0.02 % - Offset/zero point: ±0.03 % Differing values in near-range applications  A0033255 9 Maximum measured error in near-range applications; values for standard version Δ Maximum measured error R Reference point of the distance measurement D Distance from reference point of antenna
Measured value resolution	Dead band as per EN61298-2: - Digital: 1 mm (0.04 in) - Analog: 4 μA
Response time	The response time can be configured. The following step response times apply (in accordance with DIN EN 61298-2) when damping is switched off: Tank height <20 m (66 ft) Sampling rate 1 s⁻¹

Response time

<3 s

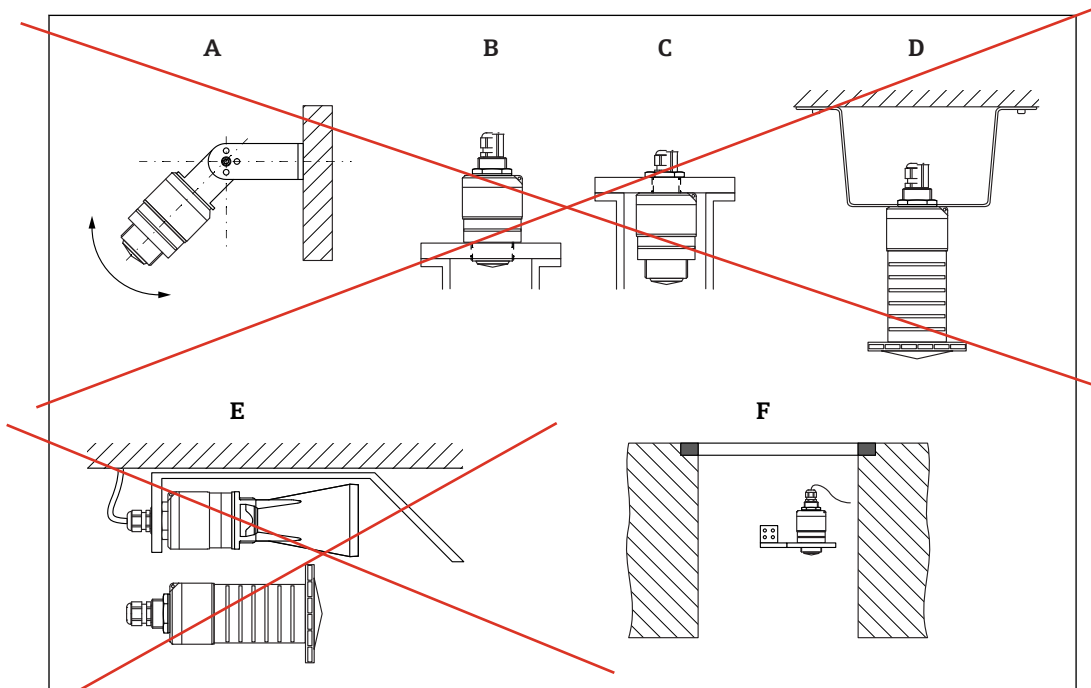


In accordance with DIN EN 61298-2, the step response time is the time following an abrupt change in the input signal up until the changed output signal has adopted 90% of the steady-state value for the first time.

Influence of ambient temperature**The measurements are carried out in accordance with EN 61298-3**

- Digital (HART, *Bluetooth*® wireless technology):
Standard version: average $T_C = \pm 3 \text{ mm } (\pm 0.12 \text{ in})/10 \text{ K}$
- Analog (current output):
 - Zero point (4 mA): average $T_K = 0.02 \text{ } \%/10 \text{ K}$
 - Span (20 mA): average $T_K = 0.05 \text{ } \%/10 \text{ K}$

Installation

Installation conditions**Installation types**

A0030605

10 Wall, ceiling or nozzle installation

- A Wall or ceiling mount, adjustable
- B Mounted at front thread
- C Mounted at rear thread
- D Ceiling installation with counter nut (included in delivery)
- E Horizontal installation in cramped spaces (wastewater shaft), 40 mm (1.5 in) antenna with flooding protection tube (accessory) or 80 mm (3 in) antenna recommended
- F Shaft wall mounting

**Caution!**

- The sensor cables are not designed as supporting cables. Do not use them for suspension purposes.
- Always operate the device in a vertical position in free-space applications.

Please note the limits for the diameter and length of the nozzle.

80 mm (3 in) antenna, installation inside nozzle

- D: min. 120 mm (4.72 in)
- L: max. 205 mm (8.07 in) + D × 4.5

80 mm (3 in) antenna, installation outside nozzle

- D: min. 80 mm (3 in)
- L: max. D × 4.5

40 mm (1.5 in) antenna, installation outside nozzle

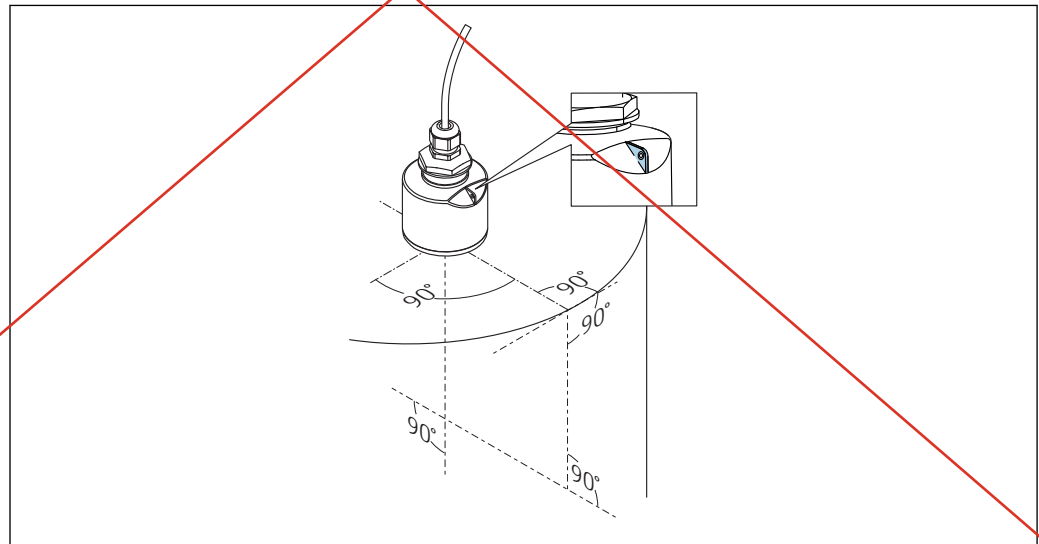
- D: min. 40 mm (1.5 in)
- L: max. D × 1.5

40 mm (1.5 in) antenna, installation inside nozzle

- D: min. 80 mm (3 in)
- L: max. 140 mm (5.5 in) + D × 1.5

Device alignment for installation on a vessel

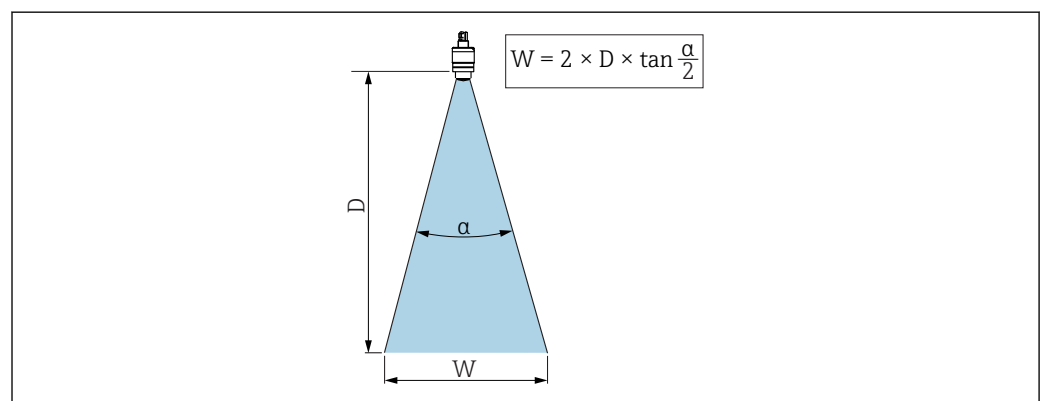
- Align the antenna vertically to the product surface.
- Align the eyelet with lug towards the vessel wall as well as possible.



A0028927

13 Device alignment for installation on a vessel

Beam angle



A0033201

14 Relationship between beam angle α , distance D and beamwidth diameter W

The beam angle is defined as the angle α , at which the power density of the radar waves reaches half the value of the maximum power density (3 dB width). Microwaves are also emitted outside the signal beam and can be reflected off interfering installations.

Beam diameter W as a function of beam angle α and distance D .

40 mm (1.5 in) antenna, $\alpha 30^\circ$

$$W = D \times 0.54$$

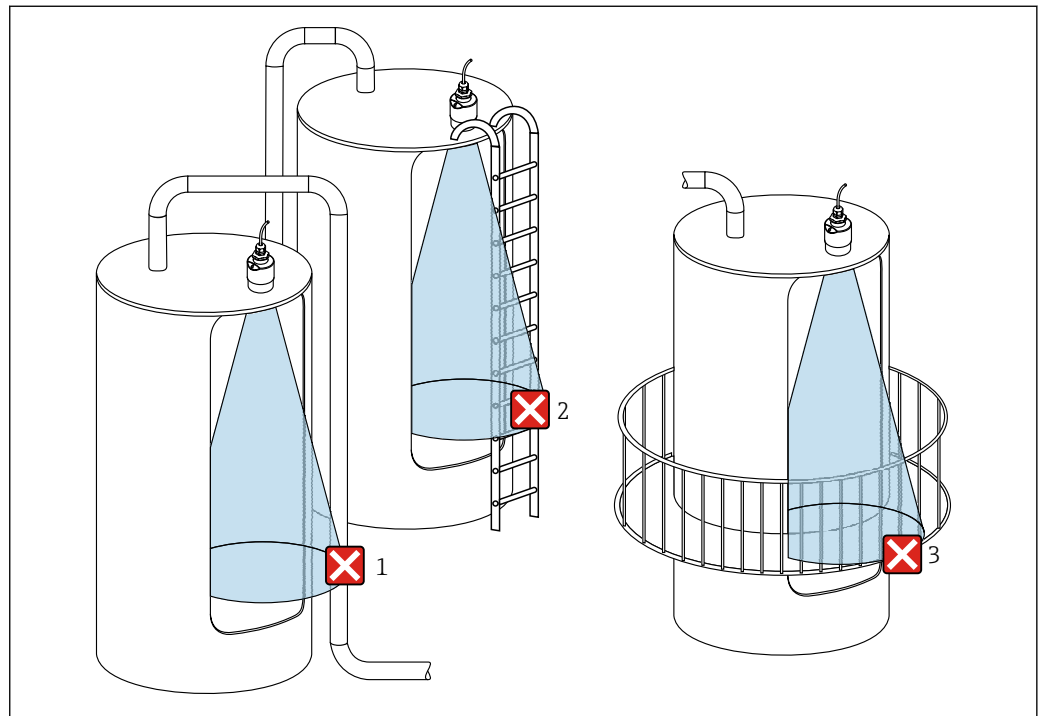
~~**40 mm (1.5 in) antenna with flooding protection tube, $\alpha 12^\circ$**~~

~~$$W = D \times 0.21$$~~

~~**80 mm (3 in) antenna with or without flooding protection tube, $\alpha 12^\circ$**~~

~~$$W = D \times 0.21$$~~

Measurement in plastic vessels



15 Measurement in a plastic vessel with a metallic, interfering installation outside of the vessel

- 1 Pipe, tubing
- 2 Ladder
- 3 Grate, railing

If the outer wall of the vessel is made of a non-conductive material (e.g. GFR), microwaves can also be reflected by interfering installations outside of the vessel.

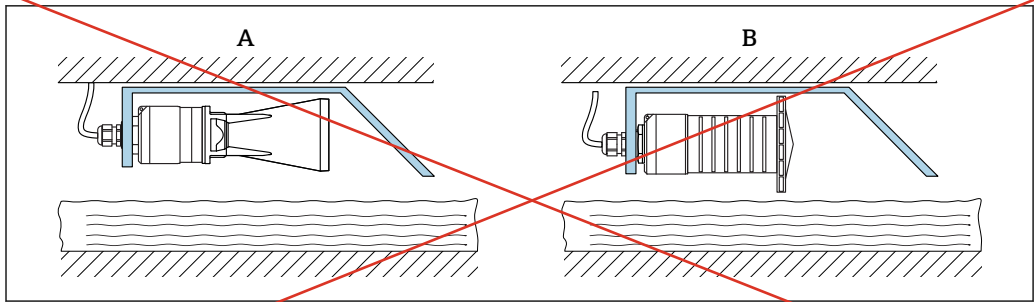
Please ensure there are no interfering installations made of a conductive material in the signal beam (see the beam angle section for information on calculating the beamwidth diameter).

Please contact the manufacturer for further information.

~~Protective hood~~

~~For outdoor use, a protective hood is recommended.~~

~~The protective hood can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".~~

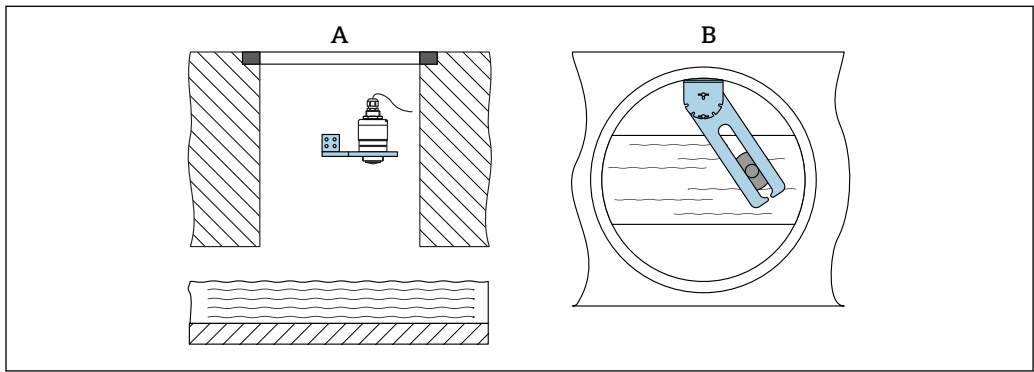


20 Installation of horizontal mounting bracket for sewer shafts

- A 40 mm (1.5 in) Antenna, with flooding protection tube (accessory)
- B 80 mm (3 in) Antenna, without flooding protection tube

Mounting in a shaft

The pivoted mounting bracket is available as an accessory.



21 Mounting in a shaft, pivotable and adjustable

- A Arm with wall bracket
- B Pivotable and adjustable arm (e.g. to align the device with the center of a channel)

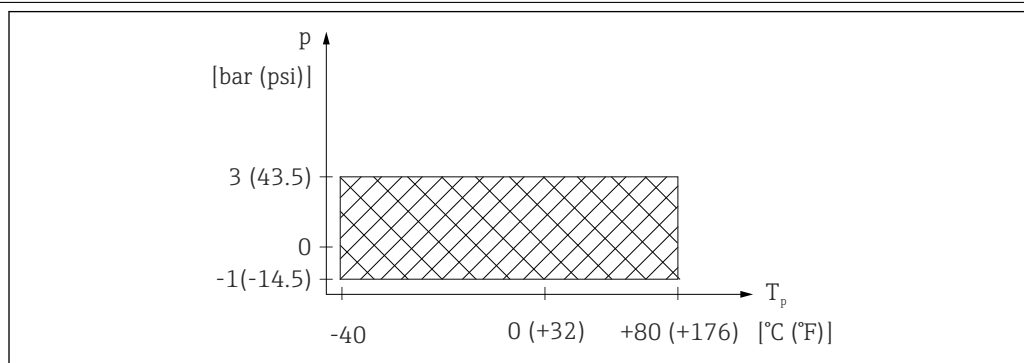
Environment

Ambient temperature range	Measuring device: -40 to +80 °C (-40 to +176 °F)  It may not be possible to use the Bluetooth connection at ambient temperatures > 60 °C (140 °F). Outdoor operation in strong sunlight: ■ Mount the device in the shade. ■ Avoid direct sunlight, particularly in warm climatic regions. ■ Use a weather protection cover.
Storage temperature	-40 to +80 °C (-40 to +176 °F)
Climate class	DIN EN 60068-2-38 (test Z/AD)
Operating altitude as per IEC 61010-1 Ed.3	Generally up to 2 000 m (6 600 ft) above sea level.
Degree of protection	Tested acc. to: ■ IP66, NEMA 4X ■ IP68, NEMA 6P (24 h at 1.83 m (6.00 ft) 1.83 m under water)
Vibration resistance	DIN EN 60068-2-64/IEC 60068-2-64: 20 to 2 000 Hz, 1 (m/s ²) ² /Hz

Electromagnetic compatibility (EMC)	Electromagnetic compatibility in accordance with all of the relevant requirements outlined in the EN 61000 series and NAMUR Recommendation EMC (NE 21). Details are provided in the Declaration of Conformity (www.endress.com/downloads).
--	---

Process

Process temperature, process pressure



A0029007-EN

22 FMR20: Permitted range for process temperature and process pressure

Process temperature range

-40 to +80 °C (-40 to +176 °F)

Process pressure range, threaded process connection

- $p_{rel} = -1$ to 3 bar (-14.5 to 43.5 psi)
- $p_{abs} < 4$ bar (58 psi)

~~Process pressure range, UNI flange process connection~~

- ~~▪ $p_{rel} = -1$ to 1 bar (-14.5 to 14.5 psi)~~
- ~~▪ $p_{abs} < 2$ bar (29 psi)~~



The pressure range may be further restricted in the event of a CRN approval.

Dielectric constant

For liquids

- $\epsilon_r \geq 4$
- Contact Endress+Hauser for lower ϵ_r values



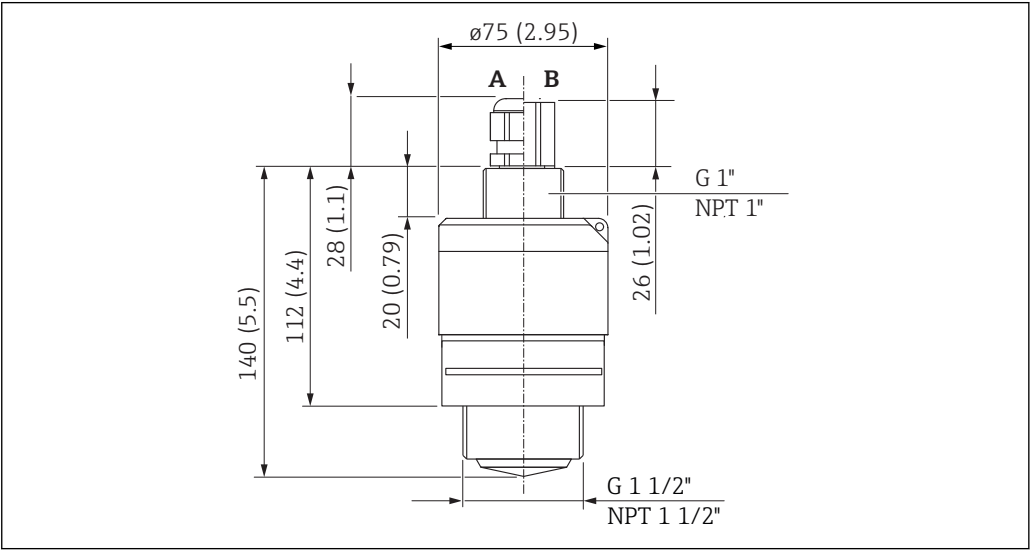
For dielectric constants (DC values) of many media commonly used in various industries refer to:

- the Endress+Hauser DC manual (CP01076F)
- the Endress+Hauser "DC Values App" (available for Android and iOS)

Mechanical construction

Dimensions

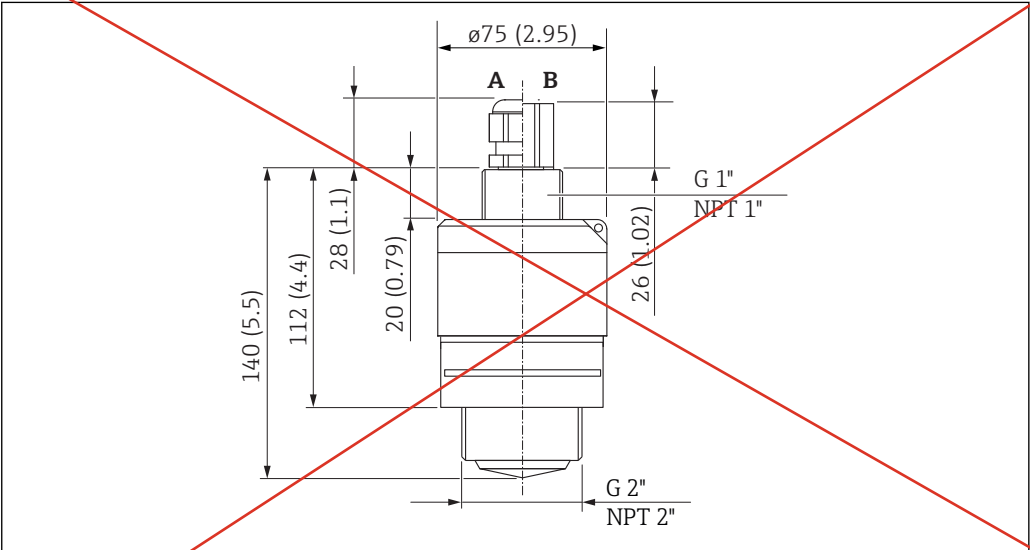
40 mm (1.5 in) Antenna with G 1-½" or MNPT 1-½" thread



23 Dimensions of G 1-½" or MNPT 1-½" process connection thread, engineering unit: mm (in)

- A Cable gland
- B FNPT ½" conduit

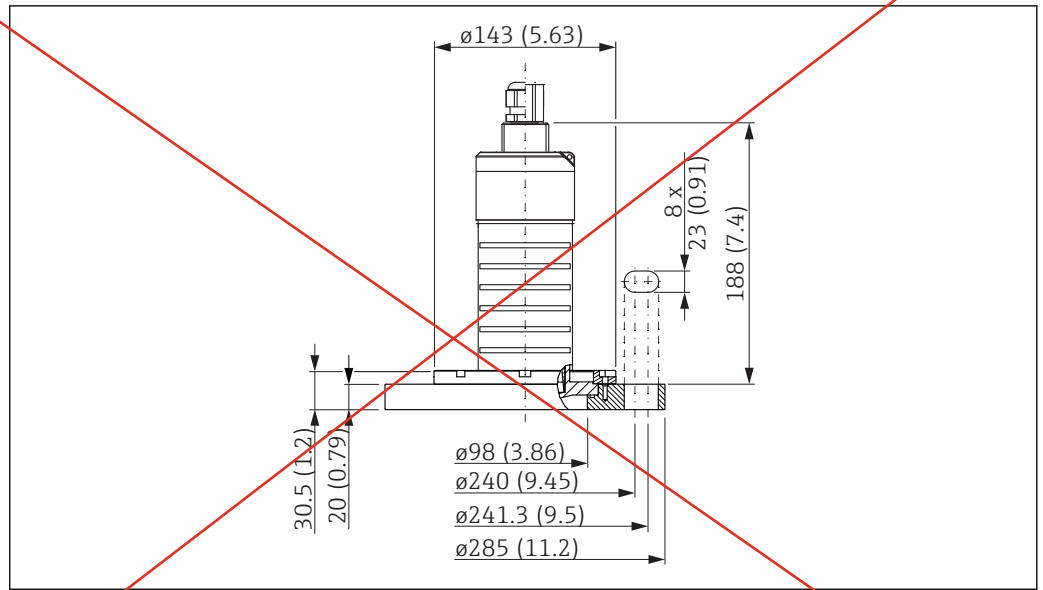
40 mm (1.5 in) Antenna with G 2" or MNPT 2" thread



24 Dimensions of G 2" or MNPT 2" process connection thread, engineering unit: mm (in)

- A Cable gland
- B FNPT ½" conduit

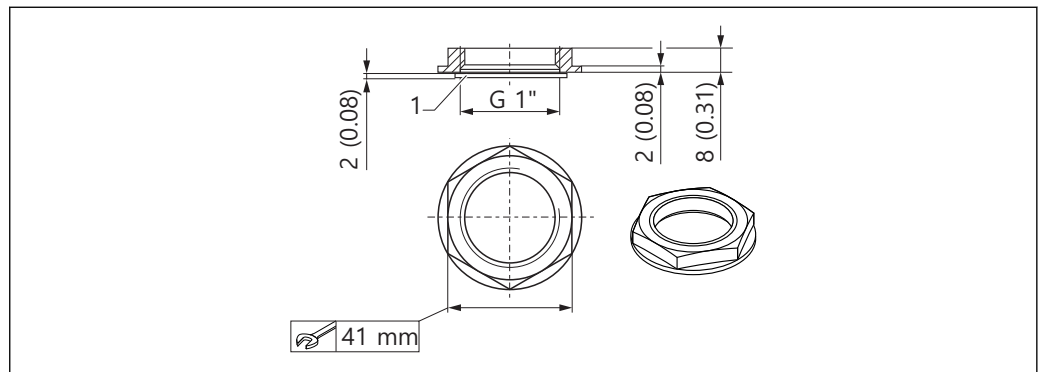
80 mm (3 in) antenna with slip-on flange 6"/DN150



30 Dimensions of 80 mm (3 in) antenna with slip-on flange 6"/DN150, engineering unit: mm (in)

The slip-on flange 6"/DN150, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

Counter nut for process connection, rear side



31 Dimensions of counter nut for process connection, rear side, engineering unit: mm (in)

1 Seal

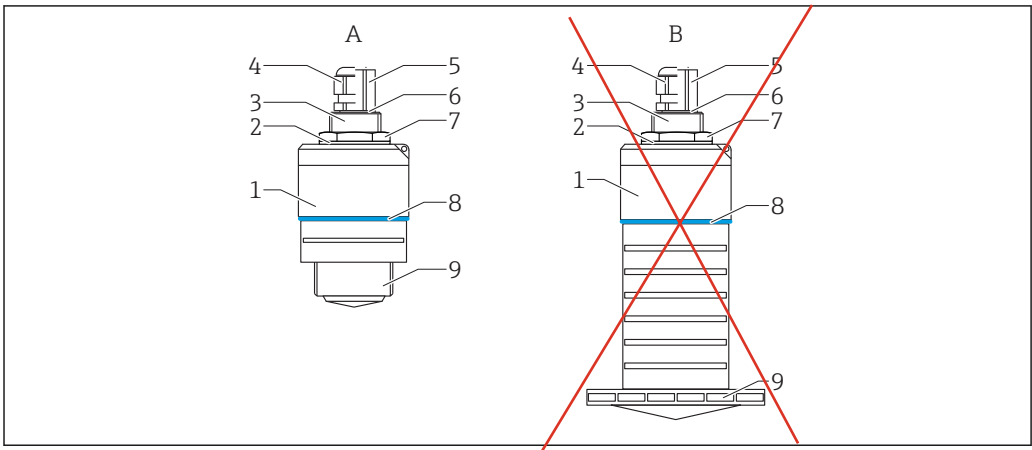
- The counter nut with seal (EPDM) is included in the scope of supply.
- Material: PA66

Weight

Weight (incl. 5 m (16.4 ft) cable)

- Device with 40 mm (1.5 in) antenna: approx. 2.5 kg (5.5 lb)
- Device with 80 mm (3 in) antenna: approx. 2.8 kg (6.2 lb)

Materials



32 Overview of materials

- A 40 mm (1.5 in) Antenna
- ~~B 80 mm (3 in) Antenna~~
- 1 Sensor housing; PVDF
- 2 Seal; EPDM
- 3 Process connection, rear side; PVDF
- 4 Cable gland; PA
- 5 Conduit adapter; CuZn nickel-plated
- 6 O-ring; EPDM
- 7 Counter nut; PA6.6
- 8 Design ring; PBT-PC
- 9 Process connection, front side; PVDF

Connecting cable

Available cable length: 5 to 300 m (16 to 980 ft)

Material : PVC

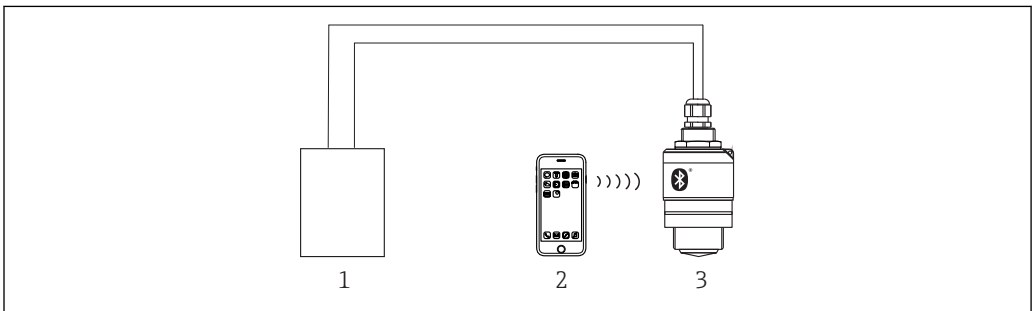
→ 20m

Operability

Operating concept

- 4 to 20 mA, HART
- Menu guidance with brief explanations of the individual parameter functions in the operating tool
- Optional: SmartBlue (app) via Bluetooth® wireless technology

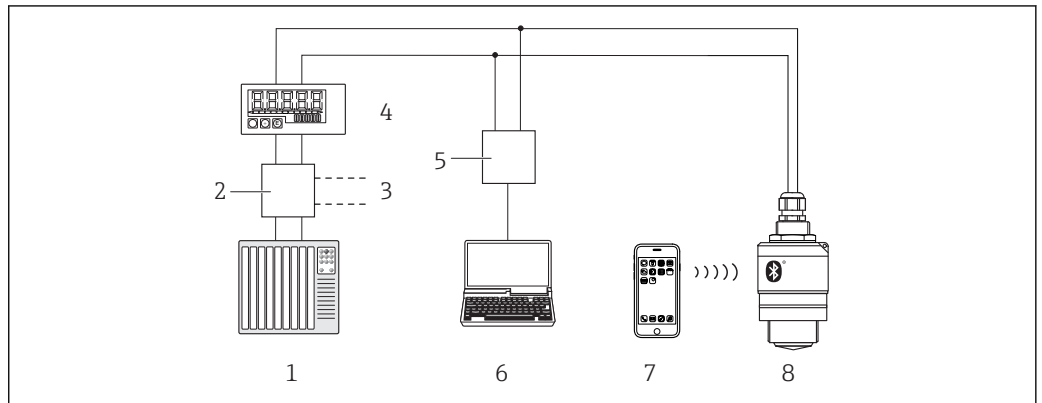
Operation via Bluetooth® wireless technology



33 Possibilities for remote operation via Bluetooth® wireless technology

- 1 Transmitter power supply unit
- 2 Smartphone/tablet with SmartBlue (app)
- 3 Transmitter with Bluetooth® wireless technology

Via HART protocol



A0028894

34 Options for remote operation via HART protocol

- 1 PLC (programmable logic controller)
- 2 Transmitter power supply unit, e.g. RN221N (with communication resistor)
- 3 Connection for Commubox FXA195
- 4 Loop-powered RIA15 process indicator
- 5 Commubox FXA195 (USB)
- 6 Computer with operating tool (FieldCare, DeviceCare)
- 7 Smartphone / tablet with SmartBlue (app)
- 8 Transmitter with Bluetooth® wireless technology

Certificates and approvals



The availability of approvals and certificates can be called up daily via the Product Configurator.

CE mark	The measuring system meets the legal requirements of the applicable EU Directives. These are listed in the corresponding EU Declaration of Conformity along with the standards applied. Endress+Hauser confirms successful testing of the device by affixing to it the CE mark.
RoHS	The measuring system complies with the substance restrictions of the Restriction on Hazardous Substances Directive 2011/65/EU (RoHS 2).
EAC conformity	The measuring system meets the legal requirements of the applicable EAC guidelines. These are listed in the corresponding EAC Declaration of Conformity together with the standards applied. Endress+Hauser confirms successful testing of the device by affixing to it the EAC mark.
RCM-Tick marking	The supplied product or measuring system meets the ACMA (Australian Communications and Media Authority) requirements for network integrity, interoperability, performance characteristics as well as health and safety regulations. Here, especially the regulatory arrangements for electromagnetic compatibility are met. The products are labelled with the RCM- Tick marking on the name plate.



A0029561

Approvals	■ Non-hazardous area ■ ATEX II 1 G Ex ia IIC T4 Ga ■ ATEX II 1/2 G Ex ia IIC T4 Ga/Gb ■ CSA C/US General Purpose ■ CSA C/US IS Cl.I Div.1 Gr.A-D, AEx ia / Ex ia T4 ■ CSA C/US Cl.I Div.2 Gr.A-D, T4 ■ EAC Ex ia IIC T4 Ga/Gb ■ IEC Ex ia IIC T4 Ga/Gb ■ KC Ex ia IIC T4 Ga/Gb ³⁾ ■ INMETRO Ex ia IIC T4 Ga/Gb ■ NEPSI Ex ia IIC T4 Ga/Gb ■ TIIS Ex ia IIC T4 ³⁾ Additional safety instructions must be followed for use in hazardous areas. Please refer to the separate "Safety Instructions" (XA) document included in the delivery. Reference to the applicable XA can be found on the nameplate.
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Explosion-protected smartphones and tablets	Only mobile end devices with Ex approval may be used in hazardous areas.
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Pressure equipment with allowable pressure ≤ 200 bar (2 900 psi)	Pressure instruments with a flange and threaded boss that do not have a pressurized housing do not fall within the scope of the Pressure Equipment Directive, irrespective of the maximum allowable pressure. Reasons: According to Article 2, point 5 of EU Directive 2014/68/EU, pressure accessories are defined as "devices with an operational function and having pressure-bearing housings". If a pressure instrument does not have a pressure-bearing housing (no identifiable pressure chamber of its own), there is no pressure accessory present within the meaning of the Directive.
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³⁾ Under development at time of going to press

harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

[Any] Changes or modifications made to this equipment not expressly approved by Endress+Hauser may void the FCC authorization to operate this equipment.

i This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

i

- The installation of the LPR/TLPR device shall be done by trained installers, in strict compliance with the manufacturer's instructions.
- The use of this device is on a "no-interference, no-protection" basis. That is, the user shall accept operations of high-powered radar in the same frequency band which may interfere with or damage this device. However, devices found to interfere with primary licensing operations will be required to be removed at the user's expense.
- Only for usage without the accessory "flooding protection tube", i.e. NOT in the free-field: This device shall be installed and operated in a completely enclosed container to prevent RF emissions, which can otherwise interfere with aeronautical navigation.

FCC / Industry Canada IDs

Tank level-probing radar

- **HVIN: FMR20**
 - FCC ID: LCGFMR2XK
 - Industry Canada ID: 2519A-2K
- **HVIN: FMR20X**
 - FCC ID: LCGFMR2XKT
 - Industry Canada ID: 2519A-2KT

Level-probing radar:

- **HVIN: FMR20+R7; FMR20+R8**
 - FCC ID: LCGFMR2XKF
 - Industry Canada ID: 2519A-2KF
- **HVIN: FMR20+R7X; FMR20+R8X**
 - FCC ID: LCGFMR2XKL
 - Industry Canada ID: 2519A-2KL

Japanese Radio Law and Japanese Telecommunications Business Law Compliance

This device is granted pursuant to the Japanese Radio Law (電波法) and the Japanese Telecommunications Business Law (電気通信事業法). This device should not be modified (otherwise the granted designation number will become invalid).

Certified No.: 202-LSF004

The products are labelled with the Technical Conformity Mark (GITEKI) from Japanese Ministry of Internal Affairs and Communications (MIC) on the name plate.



R 202-LSF004

A0032960

Mexico

El funcionamiento de este equipo está sujeto a las dos condiciones siguientes:

- (1) Este equipo o aparato no puede causar interferencias perjudiciales.
- (2) Este equipo o aparato debe aceptar todas las interferencias, incluyendo las que puedan causar un funcionamiento indeseado del equipo o aparato.

Este producto contiene un módulo inalámbrico

Marca: Endress+Hauser

Modelo: FMR20



A0034100

Other standards and guidelines

- IEC/EN 61010-1
Protection Measures for Electrical Equipment for Measurement, Control, Regulation and Laboratory Procedures
- IEC/EN 55011
"EMC Emission, RF Emission for Class B". Industrial, scientific and medical equipment – Electromagnetic disturbance characteristics - Limits and methods of measurement
- IEC/EN 61000-4-2
EMC Immunity, ESD (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Electrostatic discharge immunity test (ESD)
- IEC/EN 61000-4-3
EMC Immunity, RF field susceptibility (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test
- IEC/EN 61000-4-4
EMC Immunity, bursts (Performance Criteria B). Electromagnetic compatibility (EMC): Testing and measurement techniques - Electrical fast transient/burst immunity test
- IEC/EN 61000-4-5
EMC Immunity, surge (Performance Criteria B). Electromagnetic compatibility (EMC): Testing and measurement techniques - Surge immunity test
- IEC/EN 61000-4-6
EMC Immunity, conducted RF (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Immunity to conducted disturbances induced by radio-frequency fields
- IEC/EN 61000-4-8
EMC Immunity, magnetic fields 50 Hz. Electromagnetic compatibility (EMC): Testing and measurement techniques - Power frequency magnetic field immunity test
- EN 61000-6-3
EMC Emission, conducted RF. EMC: Radiated interference - Residential, commercial and light industry environment
- NAMUR NE 21
Electromagnetic compatibility (EMC) of industrial process and laboratory control equipment
- NAMUR NE 43
Standardization of the signal level for the breakdown information of digital transmitters with analog output signal.
- NAMUR NE 107
Status classification as per NE107
- NAMUR NE 131
Requirements for field devices for standard applications
- IEEE 802.15.1
Requirements for the *Bluetooth*® wireless technology interface

Ordering information

Detailed ordering information is available for your nearest sales organization www.addresses.endress.com or in the Product Configurator under www.endress.com :

1. Click Corporate
2. Select the country
3. Click Products
4. Select the product using the filters and search field
5. Open the product page

The Configuration button to the right of the product image opens the Product Configurator.

Product Configurator - the tool for individual product configuration

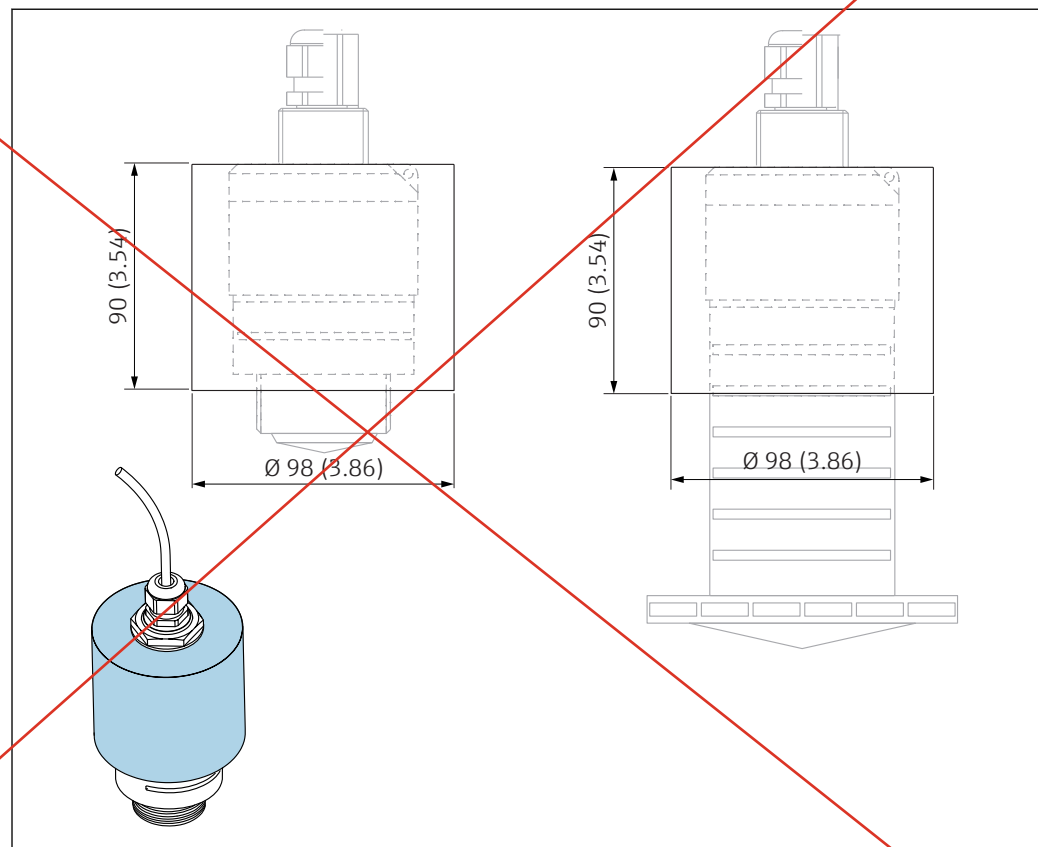
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Accessories

Device-specific accessories

Protective cover

The protective cover can be ordered together with the device via the product structure "Accessory enclosed".



 35 Dimensions of protective cover, engineering unit: mm (in)

A0028841

Material
PVDF

Data Sheet 2

Customer:	Johnston Construction
Reference:	32384 Charles County Pomonkey Pump Station Improvements
Date:	02/04/2026
<u>Qty</u>	<u>Description</u>
6	Mfg: Anchor Scientific: Float Switch, NO Model Number: GSE60NONC-GOLD Tags/Service: LSLL-01, LSL-01, LSL-02, LSH-03, LSHH-01, LSH-02 / Pump Station Level Float Switches Specifications: • G - Model G float switch • SE - External weight mounting style • 60 - 60ft cable length • NONC - 1 normally open, 1 normally closed contact • GOLD - For use in intrinsically safe applications

Float Level Switches



Listed



anchor scientific inc.



Priced From \$47.00



Priced From \$47.00

roto-float

TYPE S - SUSPENDED & TYPE P - PIPE MOUNTED

TYPES - SUSPENDED & TYPE P - PIPE MOUNTED

The ROTO-FLOAT is a direct acting float switch. Each ROTO-FLOAT contains a single pole mercury switch which actuates when the longitudinal axis of the float is horizontal, and deactuates when the liquid level falls 1" below the actuation elevation.

The float is a chemical resistant polypropylene casing with a firmly bonded electrical cable protruding. One end of the cable is permanently connected to the enclosed mercury switch and the entire assembly is encapsulated to form a completely watertight and impact resistant unit. Type S - Suspended has built-in weight.

ROTO-FLOATS can be mounted on a support pipe (type P) or suspended from above (type S). Advantages of the ROTO-FLOAT are low cost, simplicity and reliability.

NOTE: Mercury switches are not to be used in potable water.

MATERIALS OF CONSTRUCTION
Float housing.....Polypropylene
Cable clamp.....Polypropylene
Cable jacket.....P.V.C

Applications

- Pilot Duty
- Industrial Control Equipment

CABLE

- P.V.C. type STO #18 conductors (41 strand) rated 600 volts
- Various lengths available
- See table of models
- Non-standard lengths also available on special order.

Ordering information

Switch Arrangement	Cable Length	Suspended Type S Model No.	Pipe Mounted Type P Model
No.			
Normally	20	S20NO	P20NO
Open	30	S30NO	P30NO
	40	S40NO	P40NO
Normally	20	S20NC	P20NC
	30	S30NC	P30NC

eco-float

Model G

Applications

The Eco-Float can be used in a variety of liquid level monitoring applications including sumps, sewage ejectors, septic tanks, vaults, lift stations, and tanks. Eco-Floats are ruggedly constructed of corrosion resistant materials, enabling them to be used in a variety of different liquids. Some applications are subject to additional requirements described in the National Electric Code.

Description

The Eco-float is a **mercury-free** float switch for controlling liquid levels in a variety of applications. A snap-action switch is activated by a steel ball rolling back and forth within a switching tube in a plastic float housing. There is a minimum differential between "on" and "off" of approximately 3.5 inches. Greater differentials can be achieved when the pipe mounted or externally

FEATURES

- Mercury Free
- Variety of Mounting Styles
- Variety of Circuit Configurations
- Installation Easy
- Differential In One Float
- Replaces Diaphragm and Mercury Switches



Priced From \$27.50

Ordering information

Model	Mounting Style	Cable Length*	Circuit Configuration
G	SE (external weight) SI (internal weight) P (pipe-mounted)	10" 15" 20" 30" 40" 50"	NO (normally open) NC (normally closed) NONC (normally open/normally closed)

* Custom lengths available.

Example: GSI 20NO - Eco-Float, suspended internal wt. 20', normally opens contacts.

solo-float

DESCRIPTION

The new Solo-Float" is a direct acting **mercury-free** float switch for controlling 1 1/2 HP and smaller pump motors. The new Solo-Float has been completely redesigned and uses a metal ball rolling back and forth in a tube to actuate a reliable, snap-action switch. An adjustable, plastic clamp with stainless steel hose clamp is provided for attaching the Solo-Float at the appropriate level on the pump discharge pipe. By varying the tether length, a pumping range of 8" - 14" can be achieved. The Solo-Float is available with a piggyback plug in either a 115 volt or 230 volt configuration. It is also available blunt cut, without a plug, for wiring to an Anchor Scientific JX junction box.

FEATURES

- Reliability
- Adjustable Pumping Range
- Manual Operation
- Non-mercury Element
- Converts Nonautomatic Pumps To Automatic
- Replaces Diaphragm Pressure Switches
- Easy Installation
- In Stock, Low Cost



Priced From \$28.00

Ordering information

Model	Cable Length*	Circuit Configuration	Plug Voltage
D	10' 15' 20' 25' 30'	NO (normally open) NC (normally closed)	115 for 115 volt 230 for 230 volt

* Custom lengths available.

Example:

Model	Cable Length*	Circuit Configuration	Plug Voltage
D	20'	NO (normally open)	115 for 115 volt

-GOLD FOR INTRINSICALLY SAFE APPLICATION

Data Sheet 3

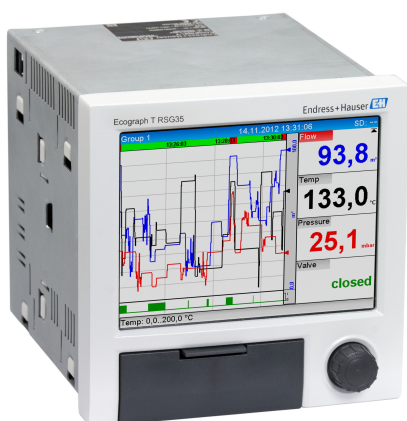
Customer:	Johnston Construction
Reference:	32384 Charles County Pomonkey Pump Station Improvements
Date:	02/04/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Endress + Hauser: Digital Chart Recorder Model Number: RSG35-B2A+C1 Tags/Service: RECORDER / PLC/RTU Panel Transmitter Digital Chart Recorder Specifications: • RSG35 - Ecograph T data manager • B - 4 universal inputs • 2 - 24 -10%...+15% AC/DC power supply • A - Ethernet RJ45 + USB communication • C1 - SD card, industrial grade, 1GB
1	Mfg: Endress + Hauser: 1GB Neutral SD Memorycard Model Number: 71187780 Tags/Service: RECORDER / PLC/RTU Panel Transmitter Digital Chart Recorder Specifications: • SD memory card • 1GB storage capacity • Generic, unbranded SD card

DIGITAL CHART RECORDER

TAGS: RECORDER

Technical Information Ecograph T, RSG35

Universal Data Manager



Record, visualize and monitor

Application

The Ecograph T graphic display recorder records and visualizes relevant process values via analog or digital input signals. The measured values are securely saved and limit values are monitored. Furthermore the Ecograph T offers intuitive operation and simple system integration. Remote configuration and visualization of the current and recorded data is easy thanks to the integrated web server - no additional software needs to be installed. In addition the Essential Version of the Field Data Manager software is also supplied with the product as standard. This software can be used to export the recorded data, save the data to an SQL database in a way that the data cannot be manipulated, and visualize the data externally.

The Ecograph T is the right solution for a wide range of applications such as:

- Quality and quantity monitoring in the water and wastewater industry
- Monitoring of processes in power stations
- Displaying and recording of critical process parameters
- Tank and level monitoring
- Temperature monitoring in metal working

Your benefits

- Versatile: up to 12 universal inputs record a wide range of measuring signals
- Clear layout: 5.7" TFT screen for displaying measured values in a maximum of four groups, with digital, bar graph and curve display
- Fast: 100 ms scan rate for all channels
- Compact: low installation depth, saves space and money
- Easy: intuitive operation via the navigator (jog/shuttle dial) on site, or user-friendly operation at the PC via the integrated web server
- Safe: reliable data archiving with internal memory and separate SD card
- Informative: e-mail notification in event of alarms and limit violation
- System-enabled: common interfaces such as Ethernet, RS232/485, USB and optional slave function for Modbus RTU/TCP
- Smart: optional mathematics functions to calculate other values
- WebDAV: data saved on SD card transmitted directly to a PC via HTTP without any additional software.

Function and system design

Measuring principle	Electronic acquisition, display, recording, analysis, remote transmission and archiving of analog and digital input signals. The device is intended for installation in a panel or cabinet. There is also the option of operating it in a desktop housing or field housing.
Measuring system	Multichannel data recording system with multicolored TFT display (145 mm / 5.7" screen size), galvanically isolated universal inputs (U, I, TC, RTD, pulse, frequency), digital inputs, transmitter power supply, limit relay, communication interfaces (USB, Ethernet, optional RS232/485), optionally available with Modbus protocol, 128 MB internal memory, external SD card and USB stick. An Essential Version of the Field Data Manager software is included for SQL-supported data analysis at the PC.  The number of inputs available in the basic device can be individually increased using a maximum of 3 plug-in cards. The device supplies power directly to connected two-wire transmitters. The device is configured and operated via the navigator (jog/shuttle dial), via the integrated web server and a PC, or via an external keyboard. Online help facilitates local operation.
Reliability	Dependability Depending on the device version, the MTBF is between 52 years and 24 years (calculated based on SN29500 standard at 40°C) Serviceability Battery-backed time and data memory. It is advisable to have the backup battery replaced by a service technician after 10 years. Real time clock (RTC) ▪ Configurable summer/normal time changeover ▪ Battery buffer. It is advisable to have the backup battery replaced by a service technician after 10 years. ▪ Drift: <10 min./year ▪ Time synchronization possible via SNTP or via digital input. Standard diagnostic functions as per Namur NE 107 The diagnostic code is made up of the error category as per Namur NE 107 and the message number. ▪ Cable open circuit, short-circuit ▪ Incorrect wiring ▪ Internal device errors ▪ Overrange/underrange detection ▪ Ambient temperature out-of-range detection Device error/alarm relay One relay can be used as an alarm relay. If the device detects a system error (e.g. hardware defect) or a malfunction (e.g. cable open circuit), the selected output/relay switches. This "alarm relay" switches if "F"-type errors occur (F = failure), i.e. "M"-type errors (M= Maintenance required) do not switch the alarm relay. Safety The tamper-proof recorded data are saved and can be transferred to an external SQL database for archiving in a way that prevents subsequent manipulation.
IT security	The manufacturer only provides a warranty if the device is installed and used as described in the Operating Instructions. The device is equipped with security mechanisms to protect it against any inadvertent changes to the device settings. IT security measures in line with operators' security standards and designed to provide additional protection for the device and device data transfer must be implemented by the operators themselves.

Input

Measured values

Number of analog universal inputs

~~Standard version without universal inputs.~~ Optional input cards (slot 1-3) with 4 universal inputs (4/8/12) each.

Number of digital inputs

6 digital inputs

Number of mathematics channels

4 mathematics channels (optional). Mathematics functions can be freely edited via a formula editor.
Integration of calculated values e.g. for totalization.

Number of limit values

30 limit values (individual channel assignment)

Function of analog universal inputs

You are free to choose between the following measured variables for each universal input: U, I, RTD, TC, pulse input or frequency input.

Integration of input variable for totalization e.g. flow rate (m³/h) in quantity (m³).

Calculated values

The values of the universal inputs can be used to perform calculations in the mathematics channels.

Measuring range of analog universal inputs

According to IEC 60873-1: An additional display error of ± 1 digit is permitted for every measured value.

User-definable measuring ranges per universal input of the multifunction card:

Measured variable	Measuring range	Maximum measured error of measuring range (oMR), temperature drift	Input resistance
Current (I)	0 to 20 mA; 0 to 20 mA quadratic 0 to 5 mA 4 to 20 mA; 4 to 20 mA quadratic ± 20 mA Overrange: up to 22 mA or -22 mA	$\pm 0.1\%$ oMR Temperature drift: $\pm 0.01\%/K$ oMR	Load: 50 Ω $\pm 1 \Omega$
Voltage (U) >1 V	0 to 10 V; 0 to 10 V quadratic 0 to 5 V 1 to 5 V; 1 to 5 V quadratic ± 10 V ± 30 V	$\pm 0.1\%$ oMR Temperature drift: $\pm 0.01\%/K$ oMR	$\geq 1 M\Omega$
Voltage (U) ≤ 1 V	0 to 1 V; 0 to 1 V quadratic ± 1 V ± 150 mV	$\pm 0.1\%$ oMR Temperature drift: $\pm 0.01\%/K$ oMR	$\geq 2.5 M\Omega$
Resistance thermometer (RTD)	Pt100: -200 to 850 °C (-328 to 1562 °F) (IEC 60751:2008, $\alpha=0.00385$) Pt100: -200 to 510 °C (-328 to 950 °F) (JIS C 1604:1984, $\alpha=0.003916$) Pt100: -200 to 850 °C (-328 to 1562 °F) (GOST 6651-94, $\alpha=0.00391$) Pt500: -200 to 850 °C (-328 to 1562 °F) (IEC 60751:2008, $\alpha=0.00385$) Pt500: -200 to 510 °C (-328 to 950 °F) (JIS C 1604:1984, $\alpha=0.003916$) Pt1000: -200 to 600 °C (-328 to 1112 °F) (IEC 60751:2008, $\alpha=0.00385$) Pt1000: -200 to 510 °C (-328 to 950 °F) (JIS C 1604:1984, $\alpha=0.003916$)	4-wire: $\pm 0.1\%$ oMR 3-wire: $\pm (0.1\% \text{ oMR} + 0.8 K)$ 2-wire: $\pm (0.1\% \text{ oMR} + 1.5 K)$ Temperature drift: $\pm 0.01\%/K$ oMR	
	Cu50: -50 to 200 °C (-58 to 392 °F) (GOST 6651-94, $\alpha=4260$) Cu50: -200 to 200 °C (-328 to 392 °F) (GOST 6651-94, $\alpha=4280$) Pt50: -200 to 1100 °C (-328 to 2012 °F) (GOST 6651-94, $\alpha=0.00391$) Cu100: -200 to 200 °C (-328 to 392 °F) (GOST 6651-94, $\alpha=4280$)	4-wire: $\pm 0.2\%$ oMR 3-wire: $\pm (0.2\% \text{ oMR} + 0.8 K)$ 2-wire: $\pm (0.2\% \text{ oMR} + 1.5 K)$ Temperature drift: $\pm 0.02\%/K$ oMR	

Measured variable	Measuring range	Maximum measured error of measuring range (oMR), temperature drift	Input resistance
	Pt46: -200 to 1100 °C (-328 to 2012 °F) (GOST 6651-94, $\alpha=0.00391$) Cu53: -200 to 200 °C (-328 to 392 °F) (GOST 6651-94, $\alpha=4280$)	4-wire: $\pm 0.3\%$ oMR 3-wire: $\pm(0.3\% \text{ oMR} + 0.8 \text{ K})$ 2-wire: $\pm(0.3\% \text{ oMR} + 1.5 \text{ K})$ Temperature drift: $\pm 0.02\%/K$ oMR	
Thermocouples (TC)	Type J (Fe-CuNi): -210 to 1200 °C (-346 to 2192 °F) (IEC 60584:2013) Type K (NiCr-Ni): -270 to 1300 °C (-454 to 2372 °F) (IEC 60584:2013) Type L (NiCr-CuNi): -200 to 800 °C (-328 to 1472 °F) (GOST R8.585:2001) Type L (Fe-CuNi): -200 to 900 °C (-328 to 1652 °F) (DIN 43710-1985) Type N (NiCrSi-NiSi): -270 to 1300 °C (-454 to 2372 °F) (IEC 60584:2013) Type T (Cu-CuNi): -270 to 400 °C (-454 to 752 °F) (IEC 60584:2013)	$\pm 0.1\%$ oMR from -100 °C (-148 °F) $\pm 0.1\%$ oMR from -130 °C (-202 °F) $\pm 0.1\%$ oMR from -100 °C (-148 °F) $\pm 0.1\%$ oMR from -100 °C (-148 °F) $\pm 0.1\%$ oMR from -100 °C (-148 °F) $\pm 0.1\%$ oMR from -200 °C (-328 °F) Temperature drift: $\pm 0.01\%/K$ oMR	$\geq 1 \text{ M}\Omega$
	Type A (W5Re-W20Re): 0 to 2500 °C (32 to 4532 °F) (ASTME 988-96) Type B (Pt30Rh-Pt6Rh): 42 to 1820 °C (107.6 to 3308 °F) (IEC 60584:2013) Type C (W5Re-W26Re): 0 to 2315 °C (32 to 4199 °F) (ASTME 988-96) Type D (W3Re-W25Re): 0 to 2315 °C (32 to 4199 °F) (ASTME 988-96) Type R (Pt13Rh-Pt): -50 to 1768 °C (-58 to 3214 °F) (IEC 60584:2013) Type S (Pt10Rh-Pt): -50 to 1768 °C (-58 to 3214 °F) (IEC 60584:2013)	$\pm 0.15\%$ oMR from 500 °C (932 °F) $\pm 0.15\%$ oMR from 600 °C (1112 °F) $\pm 0.15\%$ oMR from 500 °C (932 °F) $\pm 0.15\%$ oMR from 500 °C (932 °F) $\pm 0.15\%$ oMR from 100 °C (212 °F) $\pm 0.15\%$ oMR from 100 °C (212 °F) Temperature drift: $\pm 0.01\%/K$ oMR	$\geq 1 \text{ M}\Omega$
Pulse input (I) ¹⁾	Min. Pulse length 40 μs , max. 12.5 kHz; 0 to 7 mA = LOW; 13 to 20 mA = HIGH		Load: 50 Ω $\pm 1 \Omega$
Frequency input (I) ¹⁾	0 to 10 kHz, overrange: up to 12.5 kHz; 0 to 7 mA = LOW; 13 to 20 mA = HIGH	$\pm 0.02\%$ @ $f < 100 \text{ Hz}$ of reading $\pm 0.01\%$ @ $f \geq 100 \text{ Hz}$ of reading Temperature drift: 0.01% of measured value over the entire temperature range	

- 1) If a universal input is used as a frequency or pulse input, a series resistor must be used in series connection with the voltage source. Example: 1.2 k Ω series resistor at 24 V

Maximum load of inputs

Limit values for input voltage and current as well as cable open circuit detection/line influence/temperature compensation:

Measured variable	Limit values (steady-state, without destroying input)	Cable open circuit detection/line influence/temperature compensation
Current (I)	Maximum permitted input voltage: 2.5 V Maximum permitted input current: 50 mA	4 to 20 mA range with disengageable cable open circuit monitoring to NAMUR NE43. The following error ranges apply when NE43 is switched on: $\leq 3.8 \text{ mA}$: underrange $\geq 20.5 \text{ mA}$: overrange $\leq 3.6 \text{ mA}$ or $\geq 21.0 \text{ mA}$: open circuit (display shows: - - -)
Pulse, frequency (I)	Maximum permitted input voltage: 2.5 V Maximum permitted input current: 50 mA	No cable open circuit monitoring
Voltage (U) > 1 V	Maximum permitted input voltage: 35 V	1 to 5 V range with disengageable cable open circuit monitoring: $< 0.8 \text{ V}$ or $> 5.2 \text{ V}$: cable open circuit (display shows: - - -)
Voltage (U) $\leq 1 \text{ V}$	Maximum permitted input voltage: 24 V	
Resistance thermometer (RTD)	Measuring current: $\leq 1 \text{ mA}$	Maximum barrier resistance (or line resistance): 4-wire: max. 200 Ω ; 3-wire: max. 40 Ω Maximum influence of barrier resistance (or line resistance) for Pt100, Pt500 and Pt1000: 4-wire: 2 ppm/ Ω , 3-wire: 20 ppm/ Ω Maximum influence of barrier resistance (or line resistance) for Pt46, Pt50, Cu50, Cu53, Cu100 and Cu500: 4-wire: 6 ppm/ Ω , 3-wire: 60 ppm/ Ω Cable open circuit monitoring if any connection is interrupted.
Thermocouples (TC)	Maximum permitted input voltage: 24 V	Influence of line resistance: $< 0.001\%/ \Omega$ Error, internal temperature compensation: $\leq 2 \text{ K}$

Scan rate

Current/voltage/pulse/frequency input: 100 ms per channel

Thermocouples and resistance temperature detector: 1 s per channel

Data storage/save cycle

Selectable save cycle. Choose from: 1s / 2s / 3s / 4s / 5s / 10s / 15s / 20s / 30s / 1min / 2min / 3min / 4min / 5min / 10min / 15min / 30min / 1h

Typical recording duration

Prerequisites for following tables:

- No limit value violation / integration
- Digital input not used
- Signal analysis 1: off, 2: day, 3: month, 4: year
- No active mathematics channels



Frequent entries in the event log reduce the memory availability!

128 MB internal memory:

Analog inputs	Channels in groups	Storage cycle (weeks, days, hours)				
		5 min	1 min	30 s	10 s	1 s
1	1/0/0/0	668, 4, 14	135, 0, 5	67, 4, 4	22, 3, 20	2, 1, 18
4	4/0/0/0	491, 0, 10	99, 4, 17	49, 6, 12	16, 4, 15	1, 4, 16
8	4/4/0/0	246, 1, 14	49, 6, 1	24, 6, 19	8, 2, 7	0, 5, 20
12	4/4/4/0	164, 2, 4	33, 1, 18	16, 4, 13	5, 3, 21	0, 3, 21

External memory, 1 GB SD card:

Analog inputs	Channels in groups	Storage cycle (weeks, days, hours)				
		5 min	1 min	30 s	10 s	1 s
1	1/0/0/0	12825, 5, 20	2580, 4, 18	1291, 2, 5	430, 4, 14	43, 0, 12
4	4/0/0/0	8672, 5, 12	1749, 6, 13	875, 6, 13	292, 1, 8	29, 1, 14
8	4/4/0/0	4343, 1, 1	875, 1, 17	438, 0, 6	146, 0, 17	14, 4, 7
12	4/4/4/0	2896, 6, 13	583, 3, 21	292, 0, 6	97, 2, 20	9, 5, 4

Converter resolution

24 bit

Totalization

The interim, daily, monthly and yearly value and the total value can be determined (13-digit, 64 bit).

Analysis

Recording of quantity/operating time (standard function), also a min/max/median analysis within the set time frame.

Digital inputs	Input level	To IEC 61131-2: logical "0" (corresponds to -3 to +5 V), activation with logical "1" (corresponds to +12 to +30 V)
	Input frequency	max. 25 Hz
	Pulse length	Min. 20 ms (pulse counter)
	Pulse length	Min. 100 ms (control input, messages, operating time)
	Input current	max. 2 mA
	Input voltage	Max. 30 V

Selectable functions

- Functions of the digital input: control input, ON/OFF message, pulse counter (13-digit, 64 bit), operating time, message+operating time, quantity from time, Modbus slave.
- Functions of the control input: start recording, screen saver on, lock setup, time synchronization, limit monitoring on/off, lock keyboard/navigator, start/stop analysis.

Output**Auxiliary voltage output**

The auxiliary voltage output can be used for loop power supply or to control the digital inputs. The auxiliary voltage is short-circuit proof and galvanically isolated.

Output voltage	24 V _{DC} ±15%
Output current	Max. 250 mA

Galvanic isolation

All inputs and outputs are galvanically isolated from each other and designed for the following testing voltages:

	Relay	Digital in	Analog in	Ethernet	RS232/RS485	USB	Auxiliary voltage output
Relay	500 V _{DC}	2 kV _{DC}	2 kV _{DC}	2 kV _{DC}	2 kV _{DC}	2 kV _{DC}	2 kV _{DC}
Digital in	2 kV _{DC}	Galvanically connected	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}
Analog in	2 kV _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}
Ethernet	2 kV _{DC}	500 V _{DC}	500 V _{DC}	-	500 V _{DC}	500 V _{DC}	500 V _{DC}
RS232/RS485	2 kV _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	-	500 V _{DC}	500 V _{DC}
USB	2 kV _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	Galvanically connected	500 V _{DC}
Auxiliary voltage output	2 kV _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	500 V _{DC}	-

Relay outputs

A mix of low voltage (230 V) and safety extra low voltage (SELV circuits) is not permitted at the connections of the relay contacts.

Alarm relay

1 alarm relay with changeover contact.

Standard relay

5 relays with NO contact, e.g. for limit value messages (can be configured as NC contact).

Relay switching capacity

- Max. switching capacity: 3 A @ 30 V DC
- Max. switching capacity: 3 A @ 250 V AC
- Min. switching load: 300 mW

Switching cycles

>10⁵

Cable specification**Cable specification, spring terminals**

All connections on the rear of the device are designed as pluggable screw or spring terminal blocks with reverse polarity protection. This makes the connection very quick and easy. The spring terminals are unlocked with a slotted screwdriver (size 0).

Please note the following when connecting:

- Wire cross-section, auxiliary voltage output, digital I/O and analog I/O: max. 1.5 mm² (14 AWG) (spring terminals)
- Wire cross-section, mains: max. 2.5 mm² (13 AWG) (screw terminals)
- Wire cross-section, relays: max. 2.5 mm² (13 AWG) (spring terminals)
- Stripping length: 10 mm (0.39 in)



No ferrules must be used when connecting flexible wires to spring terminals.

Shielding and grounding

Optimum electromagnetic compatibility (EMC) can only be guaranteed if the system components and, in particular, the lines - both sensor lines and communication lines - are shielded and the shield forms as complete a cover as possible. A shielded line must be used for sensor lines that are longer than 30 m. A shield coverage of 90% is ideal. In addition, make sure not to cross sensor lines and communication lines when routing them. Connect the shield as often as possible to the reference ground to ensure optimum EMC protection for the different communication protocols and the connected sensors.

To comply with requirements, three different types of shielding are possible:

- Shielding at both ends
- Shielding at one end on the supply side with capacitance termination at the device
- Shielding at one end on the supply side

Experience shows that the best results with regard to EMC are achieved in most cases in installations with one-sided shielding on the supply side (without capacitance termination at the device).

Appropriate internal device wiring measures must be taken to allow unrestricted operation when EMC interference is present. These measures have been taken into account for this device. Operation in the event of disturbance variables as per NAMUR NE21 is thus guaranteed.

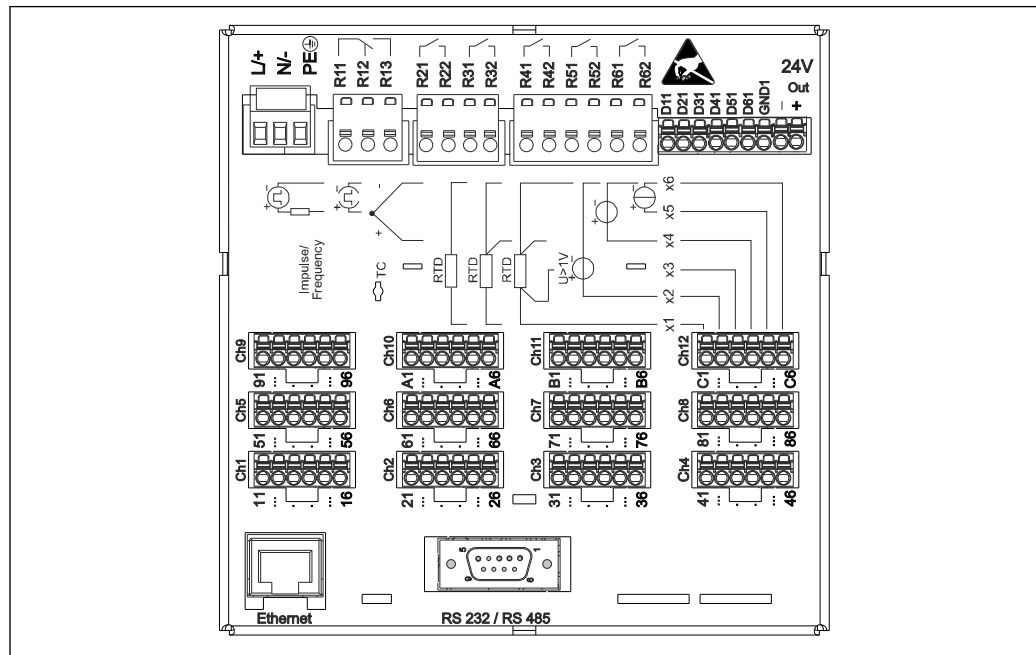
Where applicable, national installation regulations and guidelines must be observed during the installation! Where there are large differences in potential between the individual grounding points, only one point of the shielding is connected directly with the reference ground.



If the shielding of the cable is grounded at more than one point in systems without potential matching, mains frequency equalizing currents can occur. These can damage the signal cable or significantly impact signal transmission. In such cases the shielding of the signal cable is to be grounded on one side only, i.e. it may not be connected to the ground terminal of the housing. The shield that is not connected should be insulated!

Power supply

Terminal assignment



A0019304

1 Terminals on back of device

Supply voltage

- Extra-low voltage power supply unit ± 24 V AC/DC (-10% / +15%) 50/60Hz
- ~~Low-voltage power supply unit 100 to 230 V AC (+10%) 50/60Hz~~

i An overload protection element (rated current ≤ 10 A) is required for the power cable.

Power consumption

- ~~100 to 230 V: max. 35 VA~~
- 24 V: max. 24 VA

The power actually consumed depends on the individual operating state and the device version (LPS, USB, brightness of screen, number of channels, etc). The active power here is approx. 3 W to 20 W.

Power supply failure

Battery-backed time and data memory. The device starts automatically following a power failure.

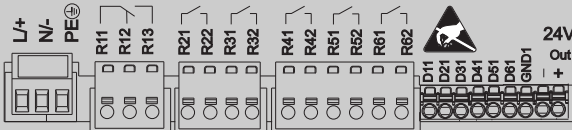
Electrical connection

Supply voltage

Power unit type	Terminal		
100-230 VAC	L+	N-	PE
	Phase L	Zero conductor N	Ground
24 V AC/DC	L+	N-	PE
	Phase L or +	Zero conductor N or -	Ground

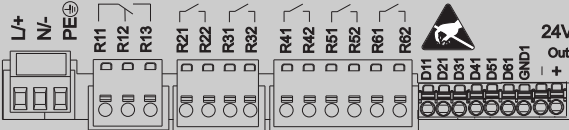
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Relay

Type	Terminal (max. 250 V, 3 A)				
					
Alarm relay 1	R11	R12	R13		
	Changeover contact	Normally closed contact (NC) ¹⁾	Normally open contact (NO) ²⁾		
Relay 2 to 6				Rx1	Rx2
				Switching contact	Normally open contact (NO ²⁾)

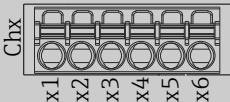
- 1) NC = normally closed (breaker)
2) NO = normally open (maker)

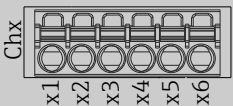
Digital inputs; auxiliary voltage output

Type	Terminal			
				
Digital input 1 to 6	D11 to D61	GND1		
	Digital input 1 to 6 (+)	Ground (-) for digital inputs 1 to 6		
Auxiliary voltage output, not stabilized, max. 250 mA			24V Out -	24V Out +
			- Ground	+ 24V (±15%)

Analog inputs

The first digit (x) of the two-digit terminal number corresponds to the associated channel:

Type	Terminal					
						
	x1	x2	x3	x4	x5	x6
Current/pulse/frequency input ¹⁾					(+)	(-)
Voltage > 1V		(+)				(-)
Voltage ≤ 1V				(+)		(-)
Resistance thermometer RTD (2-wire)	(A)					(B)

Type	Terminal					
						
Resistance thermometer RTD (3-wire)	(A)			b (sense)		(B)
Resistance thermometer RTD (4-wire)	(A)		a (sense)	b (sense)		(B)
Thermocouples TC				(+)		(-)

A0019303

- 1) If a universal input is used as a frequency or pulse input and the voltage is >2.5 V, a resistance must be used in series connection with the voltage source. Example: 1.2 kOhm series resistance at 24 V

Connector

- Panel-mounted device: connected to mains via plug-in screw terminals with reverse polarity protection
- Desktop version (option): connected to mains via IEC connector

Overvoltage protection

To avoid high-energy transients on long signal cables, connect a suitable surge arrester upstream (e.g. E+H HAW562) in series upstream.

Connection data interface, communication**USB ports (standard):**

1 x USB port type A (host)

A USB 2.0 port is available on a shielded USB A socket at the front of the device. A USB stick, for example, can be connected to this interface as a storage medium. An external keyboard or USB hub may also be connected.

1 x USB port type B (function)

A USB 2.0 port is available on a shielded USB B socket at the front of the device. This can be used to connect the device for communication with a laptop, for example.

Ethernet interface (standard):

Ethernet interface on back, 10/100 Base-T, plug type RJ45. The Ethernet interface can be used to integrate the device via a hub or switch into a PC network (TCP/ IP Ethernet). A standard patch cable (e.g. CAT5E) can be used for the connection. Using DHCP, the device can be fully integrated into an existing network without the need for additional configuration. The device can be accessed from every PC in the network. Normally only the automatic assignment of the IP address must be configured at the client. When the device is started, it can automatically retrieve the IP address, subnet mask and gateway from a DHCP server. If a DHCP is not used, these settings must be made directly in the device (depends on the network to which the device is to be connected). Two Ethernet function LEDs are located on the rear of the device.

The following functions are implemented:

- Data communication with PC software (analysis software, configuration software, OPC server)
- Web server
- WebDAV (Web-based Distributed Authoring and Versioning) is an open standard for the provisioning of files via the HTTP protocol. The data saved on the device's SD card can be read out using a PC. A web browser or dedicated WebDAV client can be selected as network drive on the PC for this purpose.

Serial RS232/RS485 interface (option):

A combined RS232/RS485 connection is available on a shielded SUB D9 socket at the rear of the device. This can be used for data transfer and to connect a modem. For communication via modem, we recommend an industrial modem with a watchdog function.

- The following baud rates are supported: 9600, 19200, 38400, 57600, 115200
- Max. cable length with a shielded cable: 2 m (6.6 ft) (RS232), or 1000 m (3281 ft) (RS485)



Only one interface can be used at any one time (RS232 or RS485).

Performance characteristics

Response time	Input	Output	Time [ms]
	Current, voltage, pulse	Relay	≤ 550
	RTD	Relay	≤ 1150
	TC ¹⁾	Relay	≤ 1550
	Cable open circuit detection, current input	Relay	≤ 1150
	Sensor error RTD, TC	Relay	≤ 5000
	Digital input	Relay	≤ 350

1) If internal measuring point temperature compensation is used, otherwise values as for voltage

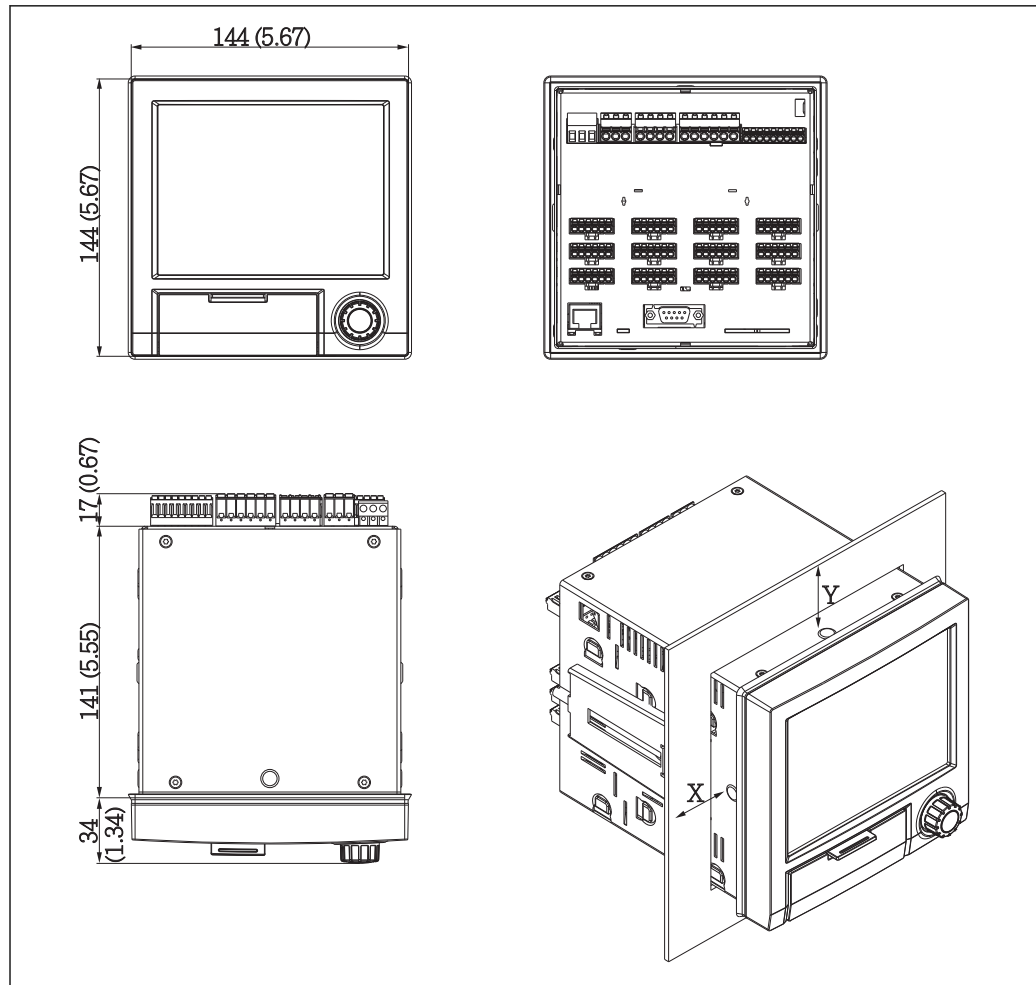
Reference operating conditions	Reference temperature	25 °C (77 °F) ±5 K
	Warm-up period	120 min.
	Humidity	20 to 60 % rel. humidity

Hysteresis Can be configured for limit values in the setup

Long-term drift As per IEC 61298-2: max. ±0.1%/year (of measuring range)

Installation

Mounting location and installation dimensions The device is designed for use in a panel in non-hazardous areas.



A0019301

■ 2 Panel mounting and dimensions in mm (in)

Please observe the installation depth of approx. 158 mm (6.22 in) for the device incl. terminals and fastening clips.

- Panel cutout: 138 to 139 mm (5.43 to 5.47 in) x 138 to 139 mm (5.43 to 5.47 in)
- Panel strength: 2 to 40 mm (0.08 to 1.58 in)
- Angle of vision: from the midpoint axis of the display, 75° to the left and right, 65° above and below.
- A minimum distance of 15 mm (0.59 in) mm (inch) between the devices must be observed if aligning the devices in the Y-direction (vertically above one another). A minimum distance of 10 mm (0.39 in) mm (inch) between the devices must be observed if aligning the devices in the X-direction (horizontally beside one another).
- Securing to DIN 43 834

Field housing assembly and design (optional)

As an option, the device can be ordered ready-mounted in a field housing with IP65.
Dimensions (B x H x D) approx.: 320 mm (12.6 in) x 320 mm (12.6 in) x 254 mm (10 in)

Desktop housing assembly and design (optional)

As an option, the device can be ordered ready-mounted in a desktop housing.
Dimensions (B x H x D) approx.: 293 mm (11.5 in) x 188 mm (7.4 in) x 211 mm (8.3 in)
(dimensions with bracket, feet and installed device)

Environment

Ambient temperature range -10 to +50 °C (14 to 122 °F)

Storage temperature	-20 to +60 °C (-4 to +140 °F)	
Humidity	5 to 85 %, non-condensing	
Climate class	To IEC 60654-1: Class B2	
Electrical safety	Class I equipment, overvoltage category II Pollution level 2	
Altitude	< 2 000 m (6 561 ft) over MSL	
Degree of protection	Front of panel-mounted device	IP65 / NEMA 4 (not assessed by UL)
	Back of panel-mounted device (terminal side)	IP20
Electromagnetic compatibility	EMC in accordance with all relevant requirements of the IEC/EN 61326 series and NAMUR NE21. For details, refer to the Declaration of Conformity. ■ Interference immunity: as per IEC/EN 61326 series (industrial environment) / NAMUR NE21 Maximum measured error <1% of measuring range ■ Interference emissions: as per IEC 61326-1 Class A	

Mechanical construction

Design, dimensions	Information about design and dimensions →  11	
Weight	■ Panel-mounted device with maximum configuration: approx. 2.2 kg (4.85 lbs) ■ Desktop housing (excluding device): approx. 2.3 kg (5 lbs) ■ Field housing (excluding device): approx. 4 kg (8.8 lbs)	
Materials	Front frame	Zinc die cast GD-Z410, powder-coated
	Sight glass	Transparent Makrolon plastic (FR clear 099) UL94-V2
	Flap; jog/shuttle dial	Plastic ABS UL94-V2
	Mounting guide rail for PCBs; motherboard fixing unit; display retainer plate	Plastic PA6-GF15 UL94-V2
	Seal to panel wall; seal to display; seal in flap; seal to navigator	Rubber EPDM 70 Shore A
	Casing; rear panel	Galvanized sheet steel St 12 ZE

 All materials are silicone-free.

Materials of desktop housing

- Housing half-panels: sheet steel, electrolytically plated (powder-coated)
- Side sections: aluminum extruded section (powder-coated)
- Section ends: colored polyamide

Display and operating elements

Operating concept

The device can be operated directly onsite, or via remote configuration with the PC via interfaces and operating tools (Web server, configuration software).

Web server

A Web server is integrated into the device. The Web server offers the following range of functions:

- Easy configuration without additional installed software
- Instantaneous value display and diagnostics information
- Display of current measured value curves via Web browser (remote control)
- Display of historical measured data in numerical format or as a curve
- Display of events and logbook entries
- Loading/saving of device configuration
- Device firmware update
- Printout of device configuration

Integrated operating instructions

The device's simple operating concept enables you to perform commissioning for many applications without the need for hard copy operating instructions. The device has an integrated help function and displays operating instructions directly on screen if the navigator (jog/shuttle dial) is pressed for longer than 3 seconds.

Local operation

Display elements

Type

TFT color display

Size (diagonal screen measurement)

145 mm (5.7")

Resolution

VGA 307,200 pixels (640 x 480 pixels)

Backlight

70,000 h half value time (= half brightness)

Number of colors

262,000 viewable colors, 256 colors used

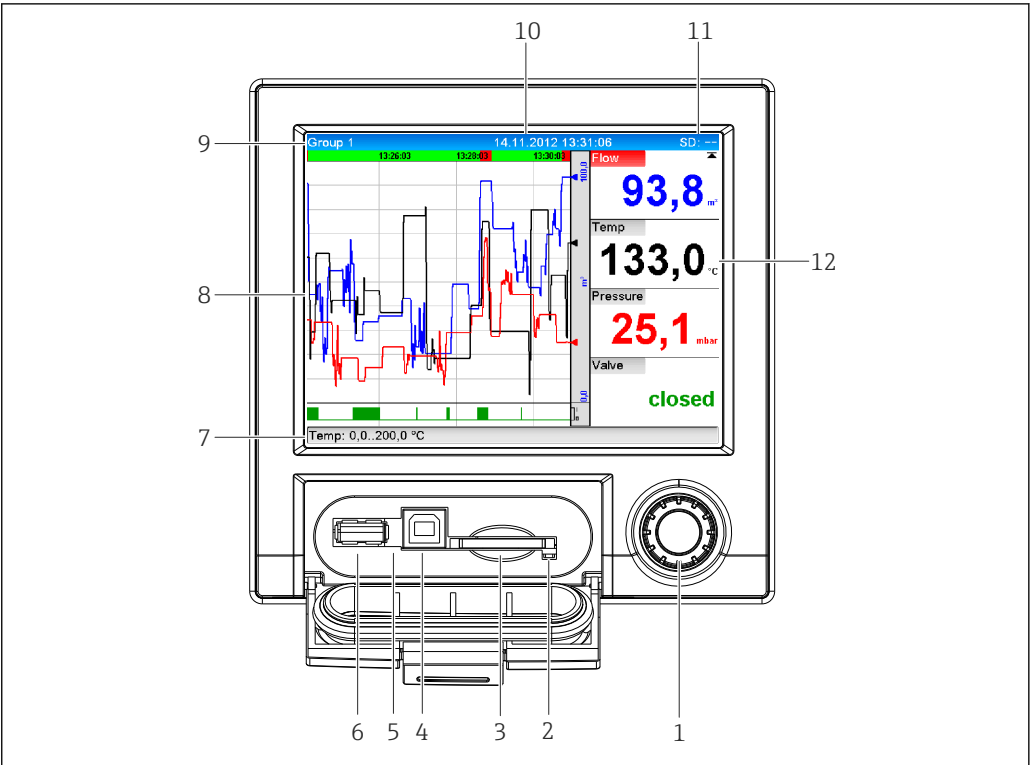
Viewing angle

Viewing angle: 130° vertical, 150° horizontal

Screen display

- Background color white
- Active channels can be assigned to up to 4 groups. These groups can be given a name e.g. "Temp. boiler 1" or "Daily averages" so that they can be uniquely identified.
- Linear scaling
- Measured value history: quick retrieval of historical data with zoom function
- Preformatted screen displays such as horizontal or vertical curve presentation, bar graph display or digital display.

Operating elements



A0020602-EN

3 Front of device with open flap

Item No.	Operating function (display mode = display of measured values) (Setup mode = operation in the Setup menu)
1	"Navigator": jog/shuttle dial for operating with additional press/hold function. In display mode: turn the dial to switch between the various signal groups. Press the dial to display the main menu. In setup mode or in a selection menu: turn the dial counterclockwise to move the bar or the cursor upwards or to the left, changes the parameter. Turning clockwise moves the bar or cursor down or clockwise, changes the parameter.
2	LED at SD slot. Orange LED lights up or flashes when the device writes to or reads from the SD card. Do not remove the SD card if the LED is lit or flashing! Risk of data loss!
3	Slot for SD card
4	USB B socket "Function" e.g. to connect to PC or laptop
5	Green LED lit: power supply present
6	USB A socket "Host" e.g. for USB memory stick or external keyboard
7	In display mode: alternating status display (e.g. set zoom range) of the analog or digital inputs in the appropriate color of the channel. In setup mode: different information can be displayed here depending on the display type.
8	In display mode: window for measured value display (e.g. curve display). In setup mode: display of operating menu
9	In display mode: current group name, type of evaluation In setup mode: name of the current operating item (dialog title)
10	In display mode: displays current date/time In setup mode: --

Item No.	Operating function (display mode = display of measured values) (Setup mode = operation in the Setup menu)
11	In display mode: alternating display indicating the percentage space on the SD card or USB stick that has already been used. Status symbols are also displayed in alternation with the memory information. In setup mode: the current "direct access" operating code is displayed
12	In display mode: display of current measured values and the status in the event of an error/alarm condition. In the case of counters, the type of counter is displayed as a symbol.  If a measuring point has limit value status, the corresponding channel identifier is highlighted in red (quick detection of limit value violations). During a limit value violation and device operation, the acquisition of measured values continues uninterrupted.

Languages

The following languages can be selected in the operating menu: German, English, Spanish, French, Italian, Dutch, Swedish, Polish, Portuguese, Czech, Russian, Japanese, Chinese (Traditional), Chinese (Simplified)

Remote operation

Device access via operating tools

Device configuration and measured value retrieval can also be done via interfaces. The following operating tools are available for this purpose:

Operating tool	Functions	Access via
"Field Data Manager (FDM)" analysis software, SQL database support (included in the delivery)	- Export of saved data (measured values, analyses, event log) - Visualization and processing of saved data (measured values, analyses, event log) - Safe archiving of exported data in a SQL database	RS232/RS485, USB, Ethernet
Web server (integrated into the device; access via browser)	- Display of current and historical data and measured value curves via the Web browser - Easy configuration without additional installed software - Remote access to device and diagnostic information	Ethernet, or Ethernet via USB
OPC server (optional)	The following instantaneous values can be provided: - Analog channels - Digital channels - Mathematics - Totalizer	RS232/RS485, USB, Ethernet
"FieldCare / Device-Care" configuration software	- Device configuration - Loading and saving device data (upload/download) - Documentation of the measuring point	USB, Ethernet

System integration

The device has (optional) fieldbus interfaces for exporting process values. Measured values and statuses can also be transmitted to the device via fieldbus. Alarms or errors in the context of data transmission are displayed depending on the bus system (e.g. status byte). The process values are transferred in the same units that are used for display at the device.

Ethernet

The following functions are implemented:

- Data communication with PC software (analysis software, configuration software, OPC server)
- Web server

~~Modbus RTU/TCP slave~~

~~The device can be connected to a Modbus system via RS485 or Ethernet interface. Up to 12 analog inputs and 6 digital inputs can be transmitted via Modbus and stored in the device~~

Certificates and approvals

CE mark	The measuring system meets the legal requirements of the applicable EC guidelines. These are listed in the corresponding EC Declaration of Conformity together with the standards applied. The manufacturer confirms successful testing of the device by affixing to it the CE mark.
UL approval	UL recognized component (see www.ul.com/database , search for Keyword "E225237")
CSA	The product meets the requirements as per "CLASS 2252 05 - Process Control Equipment"
Other standards and guidelines	■ IEC 60529: Degree of protection provided by housing (IP code) ■ IEC/EN 61010-1: Safety requirements for electrical equipment for measurement, control and laboratory use ■ IEC/EN 61326 Series: Electromagnetic compatibility (EMC requirements)

Ordering information

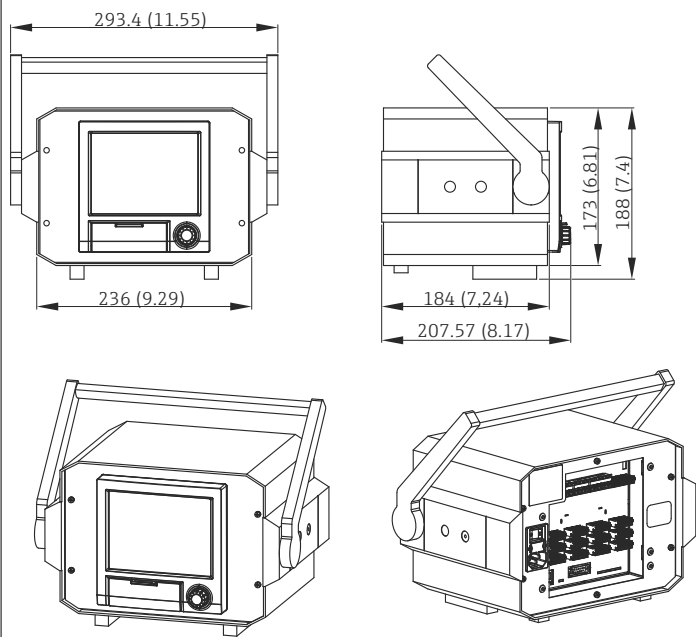
Ordering information	Detailed ordering information is available from the following sources: ■ In the Product Configurator on the Endress+Hauser website: www.endress.com -> Click "Corporate" -> Select your country -> Click "Products" -> Select the product using the filters and search field -> Open product page -> The "Configure" button to the right of the product image opens the Product Configurator. ■ From your Endress+Hauser Sales Center: www.addresses.endress.com  Product Configurator - the tool for individual product configuration ■ Up-to-the-minute configuration data ■ Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language ■ Automatic verification of exclusion criteria ■ Automatic creation of the order code and its breakdown in PDF or Excel output format ■ Ability to order directly in the Endress+Hauser Online Shop
Scope of delivery	The scope of delivery of the device comprises: ■ Device (with terminals, as per order) ■ Panel-mounted device: 2 screw fastening clips ■ USB cable ■ Optional: industrial grade SD card (card is located in the SD slot behind the flap on the front of the device) ■ "Field Data Manager (FDM)" analysis software on DVD (Essential, Demo or Professional version, depending on order) ■ Delivery note ■ Multilanguage Brief Operating Instructions, hard copy

Various accessories, which can be ordered with the device or subsequently from Endress+Hauser, are available for the device. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: www.endress.com.

Description	Order No.
"Industrial Grade" SD card, industry standard, 1GB	71213190
Field Data Manager analysis software with SQL database support (1 x workstation license, Professional version)	MS20-A1
OPC server software (full version on CD)	RXO20-11

Description	Order No.
Accessories for RXU10 data manager	RXU10- __
Designation:	
Cable set RS232 for connection to PC or modem	RXU10-B __
USB - RS232 converter	RXU10-E __
Cable USB-A - USB-B, 1.8 m (5.9 ft)	RXU10-F __
Configuration software "FieldCare Device Setup" + USB cable	RXU10-G __

Description	Order No.
Field housing IP65 (for panel-mounted device)	RXU10-H _
 4 Dimensions in mm (in)	A0021773

Description	Order No.
Desktop housing (for panel-mounted device), cable with Schuko plug Desktop housing (for panel-mounted device), cable with US plug Desktop housing (for panel-mounted device), cable with Swiss plug  5 Dimensions in mm (in)	RXU10-I _ RXU10-J _ RXU10-K _
Version: Standard Neutral	RXU10- _ 1 RXU10- _ 2

Supplementary documentation

- Standard documentation**
- Technical Information for Ecograph T RSG35: TI01079R
 - Operating Instructions for Ecograph T RSG35: BA01146R
 - Brief Operating Instructions for Ecograph T RSG35: KA01132R
 - System components and data manager - solutions to complete your measuring point: FA00016K
- Supplementary device-dependent documentation**
- Operating Instructions for Ecograph T RSG35 with Modbus RTU / TCP slave: BA01258R

Data Sheet 4

Customer:	Johnston Construction
Reference:	32384 Charles County Pomonkey Pump Station Improvements
Date:	02/04/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Precision Digital: Digital Process Meter Display Model Number: PD6000-7H2 Tags/Service: LEVEL DISPLAY / PLC/RTU Panel Process Meter Display Specifications: • 0.6" dual line red LED 6 digit SunBright display • NEMA 4X, IP65 rated with provided panel gasket • 24VDC supply voltage • -40 to 149°F operating temperature range • 4-20mA input, 2 form C SPDT relays rated at 3A at 30VDC or 3A at 250VAC resistive load • Programmable output relay operation and reset functionality • Max/Min display stored until reset by user or meter is turned off • Four front panel buttons for programming display

PD6000 ProVu Dual-Line 6-Digit Process Meter

Data Sheet



PROVu
SERIES

SUPER-BRIGHT LEDs
Our Brightest Display Ever

MeterView Pro
Instalación por USB



- 1/8 DIN Digital Panel Meter with NEMA 4X, IP65 Front
- 0-20 mA, 4-20 mA, 0-5 V, 1-5 V, and ± 10 V Field Selectable Inputs
- Dual-Line 6-Digit Display, 0.6" (15 mm) & 0.46" (12 mm)
- Isolated 24 VDC @ 200 mA Transmitter Power Supply
- ~~2 or 4 Relays with Interlocking Capability + Isolated 4-20 mA Output Options~~
- Free PC-Based MeterView Pro USB Programming Software
- No Assembly Required
- Optional SunBright Display Models for Outdoor Applications
- Operating Temperature Range: -40 to 65°C (-40 to 149°F)
- UL & C-UL Listed. E160849; 508 Industrial Control Equipment
- Input Power Options: ~~85-265 VAC~~ or 12-24 VDC
- Display Input in Two Different Scales - Great for Level Applications
- Multi-Pump Alternation Control
- Round Horizontal Tank Formula; Just Enter Diameter & Length
- 32-Point, Square Root, or Exponential Linearization
- Programmable Display, Function Keys & Digital Input
- ~~External 4-Relay & Digital I/O Expansion Modules~~
- ~~RS 232 & RS 485 Serial Communication Options with Modbus RTU~~
- ~~Light/Horn & Button Accessory~~
- ~~Control Station Accessory For Remote Operation of ProVu~~
- ~~Stainless Steel Sun Hood Accessory Available~~
- 3-Year Warranty

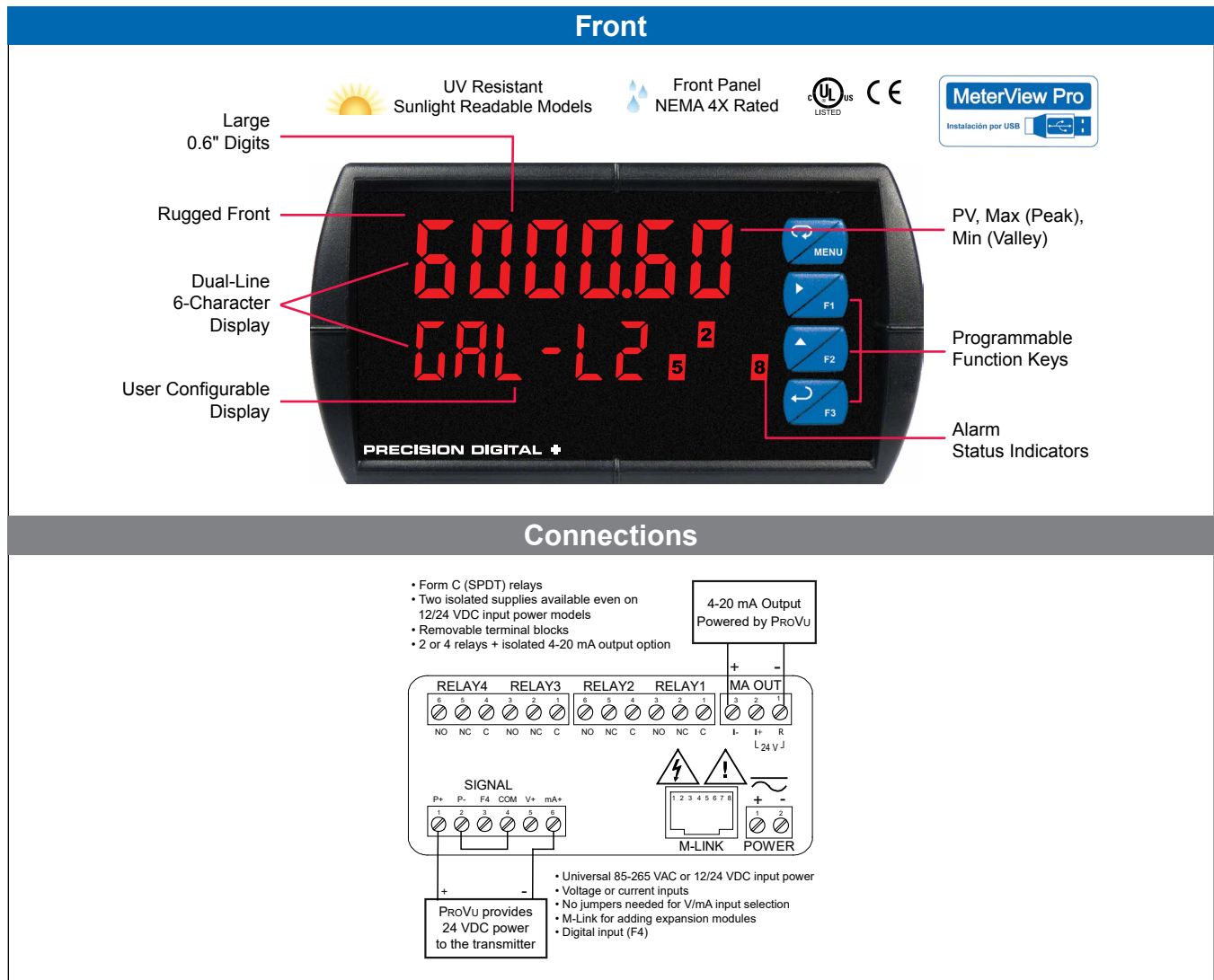


ProVu meter mounted in PDA2301 enclosure with PDA18DINSH Sun Hood and MOD-LH Light/Horn accessory. See page 16 for details.

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OVERVIEW



The Only Process Meter You Will Ever Need

Front, back and in between, the ProVu meter boasts specifications, features and functionality that make it the only 1/8 DIN process meter you will ever need. The number one feature that makes the ProVu such a useful device is its built-in 24 VDC power supply to drive the transmitter as illustrated by the above diagram. This feature not only saves the cost of an external power supply, but also greatly simplifies wiring. In addition, there is a second 40 mA power supply provided with the 4-20 mA output option, evident also in the above diagram.

The picture above illustrates several other reasons why the ProVu is the only process meter you will ever need. First off, is the NEMA 4X rated front panel which means you can install

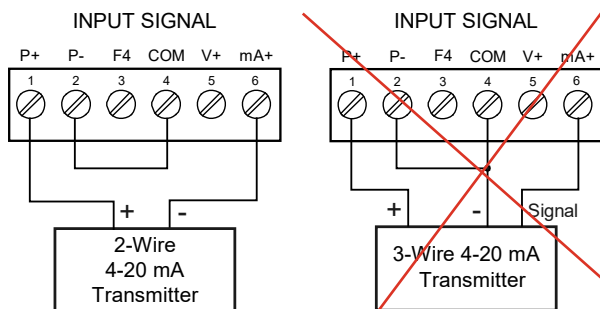
the ProVu in panels exposed to moisture, dust and other adverse conditions. The picture also points out that the ProVu is available with an optional Sunbright display which means you can install and read the ProVu in direct sunlight. The next thing to notice is the 6-digit dual-line display that can display numbers up to 999,999 on the upper line and show either a tag or the input in a different scale on the lower line.

Other key features include four relays and 4-20 mA output option, advanced signal input conditioning like automatic round horizontal tank linearization, function keys, pump alternation capability, and Modbus RTU serial communications. Finally all these features and capabilities can easily be programmed with free MeterView Pro PC-based software.

ISOLATED TRANSMITTER POWER SUPPLIES

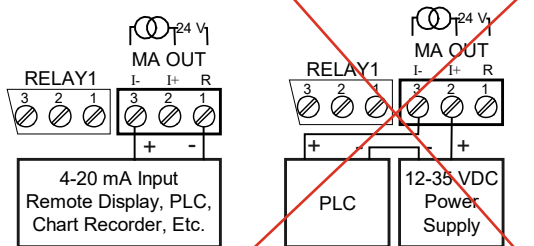
24 V @ 200 mA Transmitter Power Supply

One of the most useful standard features of the AC powered PD6000 is its built-in isolated, 24 V @ 200 mA power supply to power the transmitter. This feature saves money by eliminating an external power supply and also simplifies wiring by reducing the number of devices in the loop. It can be configured for 5, 10, or 24 V (default) by means of a simple internal jumper. This power supply is even available on meters that are powered from DC power (24 V @ 100 mA). To use an external power supply instead of the internal power supply, simply make connections to different terminals on the PROVu. The following diagrams illustrate how to wire the PROVu so it will power the transmitter:



24 V @ 40 mA 4-20 mA Output Power Supply

Not only can the PROVu power the 4-20 mA input signal, but an additional power supply of 24 V @ 40 mA is provided with the 4-20 mA output option to power the 4-20 mA output.

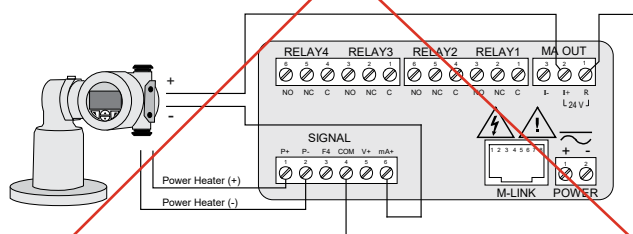


Resettable Fuse Prevents Current Overload

Another very useful aspect of the PROVu is that the current input is protected against current overload by a resettable fuse. The fuse limits the current to a safe level when it detects a fault condition, and automatically resets itself when the fault condition is removed.

Other Uses for the Transmitter Power Supplies

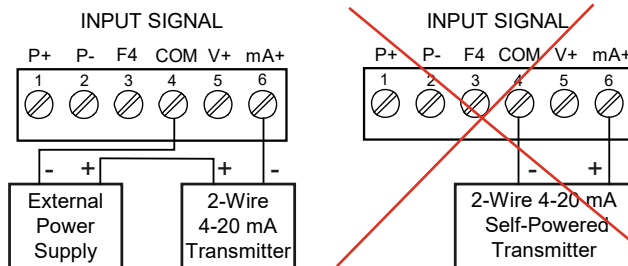
The most common use for these two power supplies is for the 200 mA transmitter power supply to power the field transmitter and 40 mA power supply to power the 4-20 mA output from the meter. However, since these two power supplies are isolated they can be used in other ways. For instance, some level transmitters require the use of a heated lens. The PROVu's 200 mA power supply could be used to power the heated lens and PROVu's 40 mA power supply could be used to power the 4-20 mA input.



PROVu Powers Both Heater and 4-20 Output

External Power Supply for the Loop

For applications that require an external transmitter power supply, the same PROVu is used and merely wired in a different fashion as the following diagrams illustrate:



PDA1024-01 24 VDC Transmitter Power Supply

Precision Digital offers the PDA1024-01 for applications that require more than the 200 mA power that the PROVu can provide.



Specifications

Output Voltage: 24 VDC $\pm 10\%$ @ 1.5A rated current

Dimensions: 1.40" x 3.50" x 2.10"

(35 mm x 90 mm x 54.5 mm) (W x H x D)

ADVANCED DISPLAY FEATURES

Dual-Line Makes All the Difference

The upper display can be programmed to indicate PV, maximum (peak), minimum (valley), alternating maximum/minimum, one of eight alarm set points, or Modbus input. The lower display can be configured to display engineering units, set points, user defined messages, or simply turned off.

The PROVu's dual-line display makes all the difference both when programming the instrument and when using it in the field. When programming the instrument, the dual line display prompts for the needed information and also helps you keep track of where you are in the setup process. When using the instrument, the dual line display provides more information such as displaying the input in two different scales like height and volume for a level application. We call this the Dual-Scale feature.

Programming Assistance

The PROVu's dual-line display makes programming the instrument much easier because the lower line prompts for the needed information and also helps you keep track of where you are in the setup process.



The PROVu is prompting for the value for Input 2 and displaying the default value of 20.00 mA. The "2" in 20.00 is brighter than the rest of the digits indicating that it is the number that will be changed by the Up and Right arrows.



The PROVu is now prompting for what the user wants Display 2 to be; that is the value that corresponds to 20 mA. In this case Display 2 is currently set to 95.00.

Bright & Optional Super-Bright Display

The standard PROVu's display is bright enough for most applications, including moderate sun exposure. However, for direct sunlight exposure the PROVu is available with super-bright LEDs that make it possible to read the PROVu even in direct sunlight. Both versions of the PROVu have eight levels of adjustable intensity.

Rounding Feature for Even Steadier Display

The rounding feature is used to give the user a steadier display with fluctuating signals. It causes the display to round to the nearest value according to the rounding value selected (1, 2, 5, 10, 20, 50, or 100). For example, with a rounding value of 10, and an input of 12346, the display would indicate 12350.

Dual-Scale Display Feature

The PROVu PD6000 has a rather unique, and very flexible dual-scale capability. This is of particular value in level applications where a second scaled display can represent the measured input in a different form (i.e. gallons & height). Both displays are independently scaled and are based on the 4-20 mA input signal. Beyond level, this function has been used for pressure & force, current & power, feet & meters, GPM & CFM, and more.



Gallons & mA



Gallons & Height

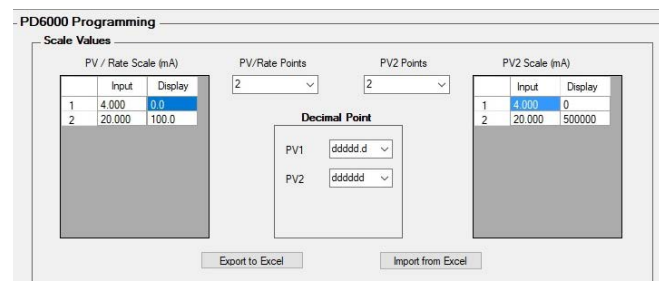


Gallons & Percent



Gallons & Head PSI

MeterView Pro can be used to program the PROVu to display the input in two different scales:



Toggle Between Reading & Units with Tag

The PROVu can also be programmed so the upper line toggles between the reading and units and the lower line displays a tag. For instance, the upper line toggles between 9500 and Gal and the lower line displays Tank 1.



Other Uses for Lower Line

The lower line can also be used indicate units, a tag, or even a setpoint as the following pictures illustrate:



Volume in Tank



Gallons & Setpoint



Level in Feet

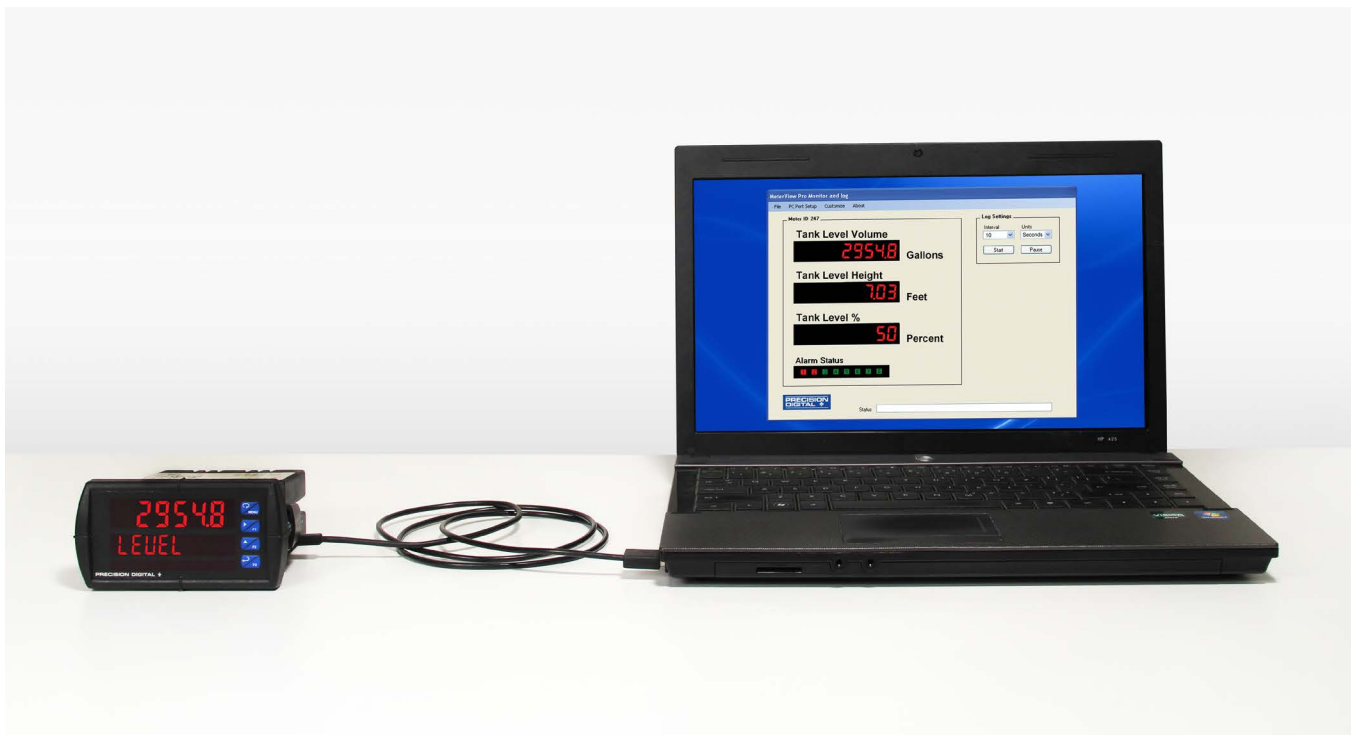


Pressure Indication

QUICK & EASY SCALE & PROGRAMMING METHODS

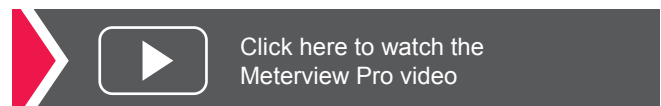
The PROVu can be programmed either via the front panel push buttons or free, PC-based MeterView Pro software. MeterView Pro is resident on the PROVu and is accessed by a provided USB cable, so it is by far the easiest way to program the PROVu. The PROVu can be calibrated either by applying a known signal or scaled by entering a desired value with the front panel buttons or MeterView Pro software. Most customers will use the scaling method because it is simpler and does not require a calibrated signal source. Selecting the input to be current or voltage is done with the front panel buttons or MeterView Pro software. Once programming is completed it can be locked with a password.

Free PC-Based MeterView Pro USB Programming Software & Cable



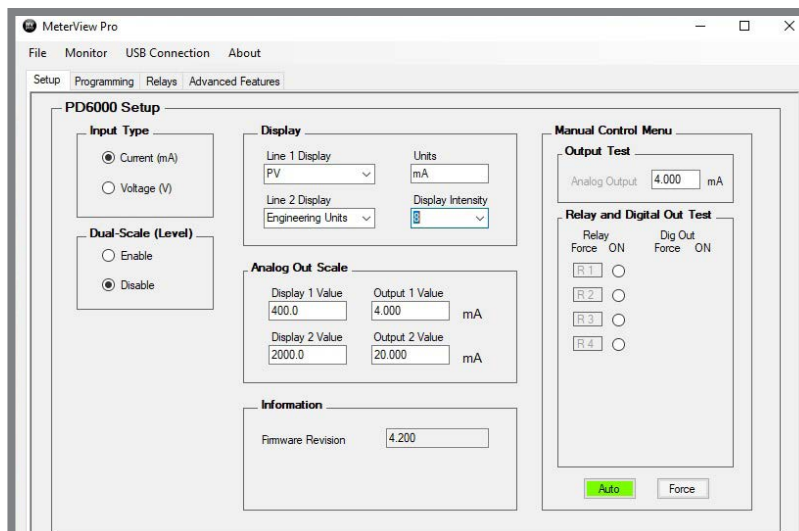
The PROVu comes preloaded with free MeterView Pro programming software that connects and installs directly to your PC with a standard USB cable, also provided free with each instrument. This eliminates the need to insert CDs, install drivers, or download software from the internet. When you connect your PROVu to your PC, MeterView Pro is downloaded to your PC, the software automatically selects the model you are programming, and you're ready to start programming immediately. Further simplifying the programming process, the

PROVu can be powered from the USB port, so no need to apply external power while programming your meter. In addition to programming, the software will also allow you to monitor, and datalog a PROVu using your PC. You can also generate and save programming files for later use.



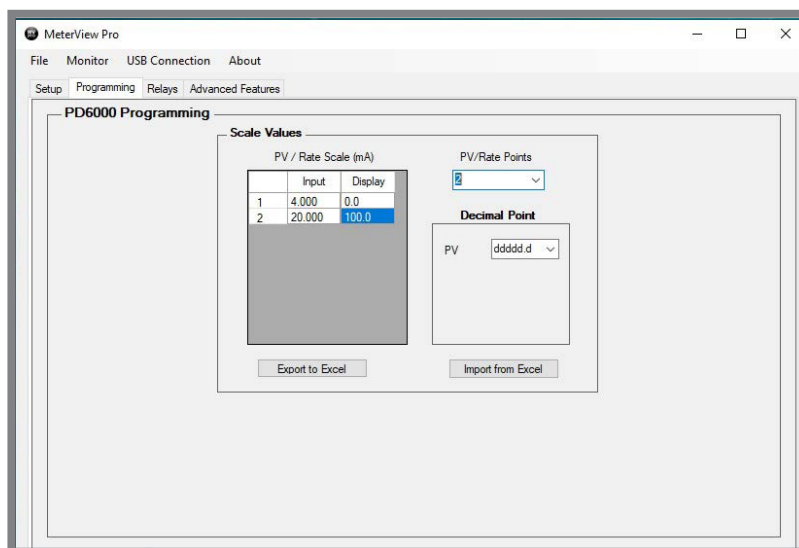
Setup Screen

- Select Voltage or Current Input
- Activate Dual-Scale (Level) Function
- Select Decimal Point
- Set Line 1 Display Parameters
- Set Line 2 Display Parameters
- Set Analog Output Values
- Enable Manual Control
- Test Relays & Digital Outputs



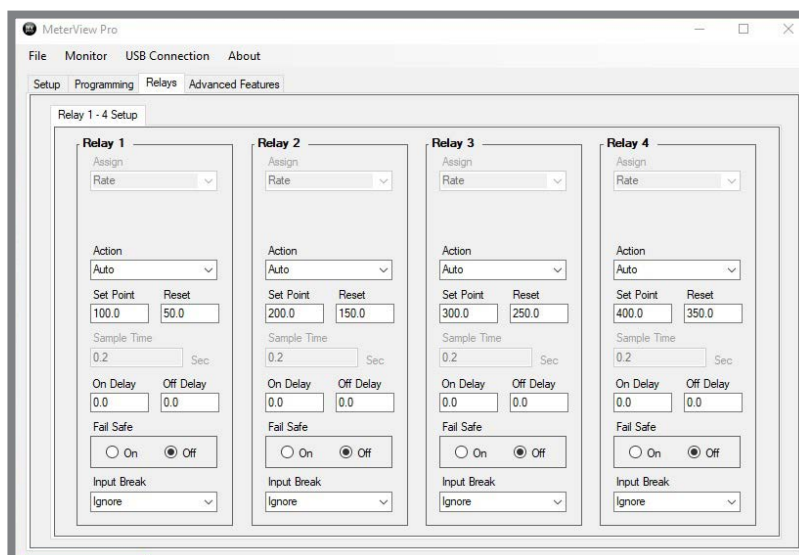
Programming Screen

- Set Scale Values
- Set the Number of Points (up to 32)
- Import from Excel
- Export to Excel



Relays Screen

- Greatly Simplifies Programming a Variety of Relay Features
- Set Relay Action
- Set Sampling Time
- Set Set & Reset Points
- Set On/Off Time Delays
- Set Fail Safe Operation
- Set Input Break Relay Action



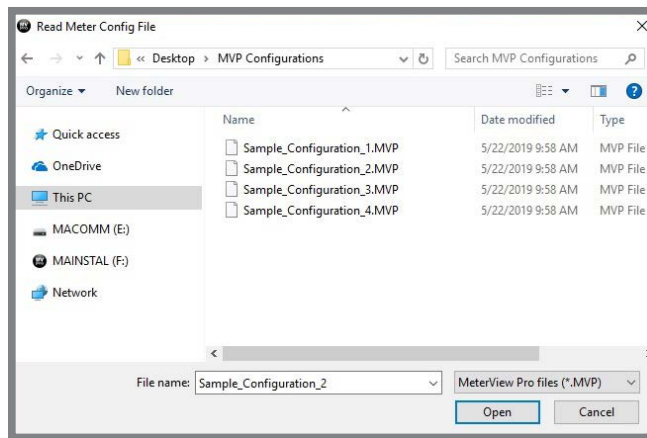
Save/Open Configuration

At the bottom of most MeterView screens are two tabs:

1. **Get Meter Data:** This reads the programming of the meter that is currently connected to the PC.
2. **Send Meter Data:** Clicking this button, sends current MeterView programming to the meter.



The configuration file can be sent or retrieved from the directory of your choice. This makes it very easy to program multiple meters with the same programming. It is also a great backup utility as well.



Specifications

System Requirements:

Microsoft® Windows® 7/10/11

Communications:

Onboard USB (firmware version 4.0 or higher),
RS-232 Adapter or RS-485 Adapter

Meter Address: 1 - 247

Reports:

- Data logging: Save as CSV file format
- Configuration: Save as PDC file format or print configuration

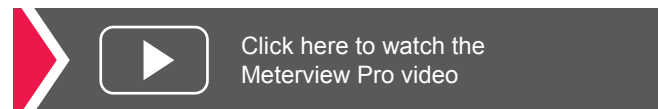
Baud Rate: 300 - 19,200 bps

Configuration: One meter at a time

Protocol:

Modbus RTU (requires firmware version 4.0 or higher)

**Note: Windows® 32/64-bit operating systems*



Password Protection

The Password menu is used for programming three levels of security to prevent unauthorized changes to the programmed parameter settings:

Pass 1: Allows use of function keys and digital inputs

Pass 2: Allows use of function keys, digital inputs and editing set/reset points

Pass 3: Restricts all programming, function keys, and digital inputs

4-20 mA OUTPUT & RELAYS

4-20 mA Analog Output

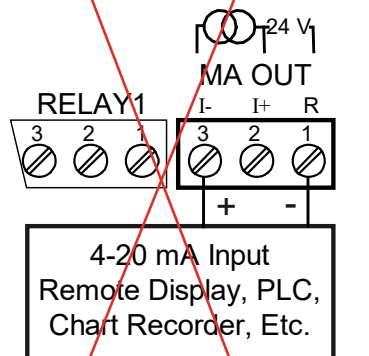
The isolated analog retransmission signal can be configured to represent the process variable (PV), maximum (peak) value, minimum (valley) value, the value for any of the eight relay set points, or Modbus input. While the output is nominally 4-20 mA, the signal will accurately accommodate under- and over-ranges from 1 to 23 mA.

The 4-20 mA output can be reversed scaled such that 4 mA represents the high value and 20 mA represents the low value. For instance, a 4-20 mA output signal could be generated as the meter went from 100.0 to 0.0.

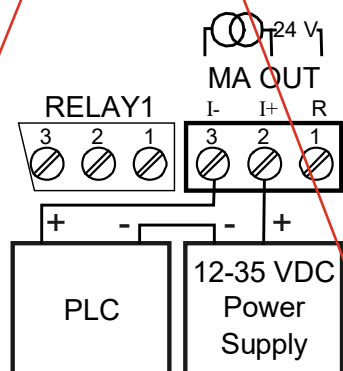
For applications where the input was linearized by the PROVu, the 4-20 mA output will represent that linearized value.

Connections

The PROVu can provide 40 mA at 24 VDC to power the 4-20 mA output signal or an external power supply can be used:



4-20 mA Output Powered by PD6000



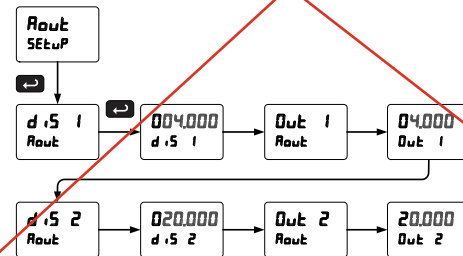
4-20 mA Output Powered by External Power Supply

The internal 24 VDC power supply powering the analog output may be used to power other devices, if the analog output is not used. The I+ terminal is the +24 V and the R terminal is the return.

The 4-20 mA output can either be programmed using the front panel push buttons or free MeterView Pro software.

Front Panel Push Button Programming

The 4-20 mA analog output can be scaled to provide a 4-20 mA signal for any display range selected. No equipment is needed to scale the analog output; simply program the display values to the corresponding mA output signal. The Analog Output menu is used to program the 4-20 mA output based on display values.



MeterView Pro Software Programming

When a meter is programmed as shown below, the output will be 4.00 mA when the display reads 0 and the output will be 20.00 mA when the display reads 150000.

Analog Out Scale	
Display 1 Value 0	Output 1 Value 4.000 mA
Display 2 Value 150000	Output 2 Value 20.000 mA

The meter can be set up for reverse scaling as shown below: the output will be 4.00 mA when the display reads 150000 and the output will be 20.00 mA when the display reads 0.

Analog Out Scale	
Display 1 Value 150000	Output 1 Value 4.000 mA
Display 2 Value 0.00	Output 2 Value 20.000 mA

Analog Out Control	
Source PV Value	
Break ☒	1.000
O-Range	21.000
U-Range	3.000
Max Output	23.000
Min Output	1.000

Source: Source for generating the 4-20 mA output (e.g. PV)

Overrange: Analog output value with display in overrange condition

Underrange: Analog output value with display in underrange condition

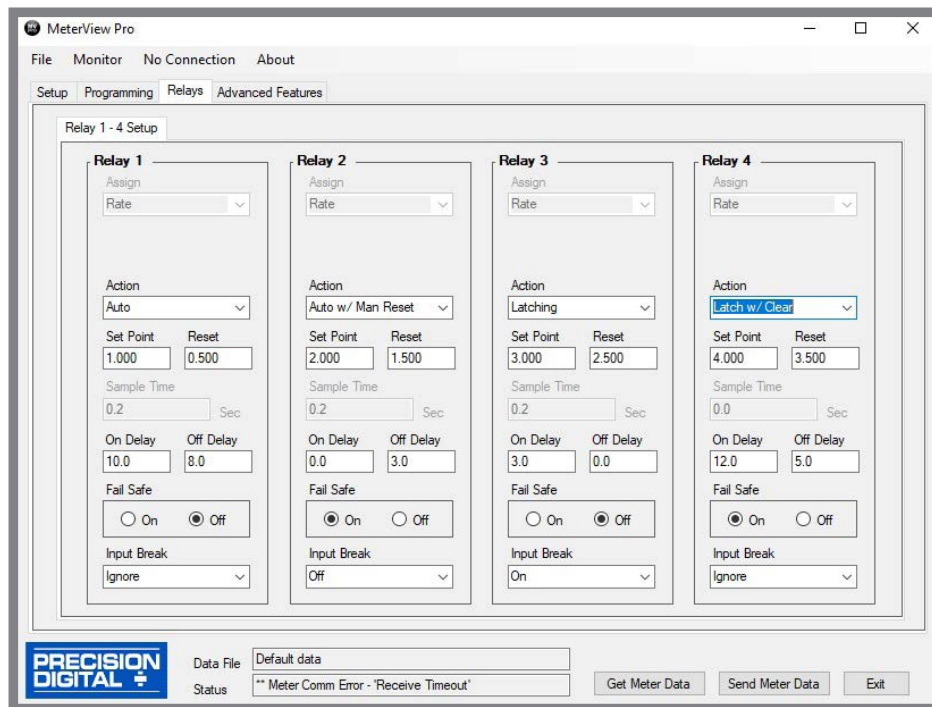
Break: Analog output value when loop break is detected

Max: Maximum analog output value allowed regardless of input

Min: Minimum analog output value allowed regardless of input

Relays for Alarm & Control Applications

Adding relays to the PROVu meter turns it into a sophisticated alarm device as well as a powerful, yet simple, alternative to a more complicated PLC system for control applications. One such application would be pump control using the PROVu's relays in pump alternation mode. The PROVu can be equipped with up to four 3 A Form C (SPDT) internal relays and an additional four more 3 A Form A (SPST) external relays. Relays are highly user-configurable as the following screen shot from MeterView Pro indicates:



**Values are intended to show programming choices. They are not intended to represent an actual application.*

Setting Set and Reset Points (HI / LO Alarms)

All relays are independent of each other and may be programmed as high or low alarms with user desired set and reset points. Setting a set point above a reset point results in a high alarm and setting a set point below a reset point results in a low alarm. Alarms have 0 – 100% deadband and set and reset points may be set anywhere in the range of the meter.

Resetting the Relays (Action in MV Pro)

All relays are independent of each other and may be programmed to reset (*Action* in MV Pro) in the following ways:

- **Automatic:** Alarm will reset automatically once the alarm condition has cleared.
- **Automatic/Manual:** Alarm will reset automatically once the alarm condition has cleared but can also be reset using the F3 front panel button* at any time.
- **Latching:** Alarm must be reset manually and can be done so at any time. Press the F3 front panel button* at any time to clear the alarm.
- **Latching with Reset after Cleared:** Alarm must be reset manually and can only be done so after the alarm condition has cleared. Press the F3 front panel button* after the alarm condition has cleared to reset the alarm.

* Or by connecting an external switch to F4 terminal or with an optional digital input.

Time Delay (On and Off)

In many applications it is desirable to wait before turning off or on a relay – such as waiting for a process to settle before taking action. Each relay on the PROVu can be programmed with independent on and off time delays of 0 to 999.9 seconds to achieve this.

Relays Auto Initialization

When power is applied to the meter, the front panel LEDs and alarm relays will reflect the state of the input to the meter.

Signal Loss or Loop Break Relay Operation

When the meter detects a break in the 4-20 mA loop, the relay will go to one of the following selected actions:

1. Turn On (Go to alarm condition)
2. Turn Off (Go to non-alarm condition)
3. Ignore (Processed as a low signal condition)

User Selectable Fail-Safe Operation

All relays are independent of each other and may be programmed for user selectable fail-safe operation. With the fail-safe feature activated, the relays will transfer to the alarm state on power loss to the meter.

Front Panel LEDs

The meter is supplied with four alarm points that include front panel LEDs to indicate alarm conditions. This standard feature is particularly useful for alarm applications that require visual-only indication.

Manual Output Control

Take control of any output with this feature. All relays can be forced ON or OFF, and the 4-20 mA output signal can be set to any value within its range. When the relays and 4-20 mA output are controlled manually, an LED labeled “M” is turned on and the associated Alarm LEDs (1-8) flash every 10 seconds indicating that the meter is in manual control mode.

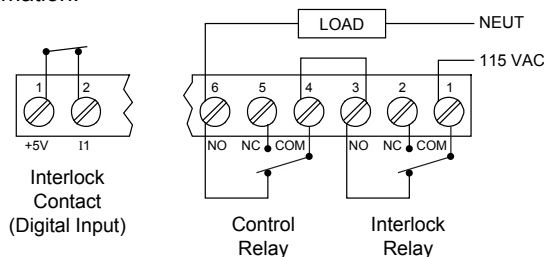


Sampling Function (PV Triggered Timed Relay)

The sampling function allows the operator to set a relay as a “sampling” relay. When the PV reaches that set point, it will close that relay’s contacts for a preset period of time (0.1 to 5999.9 seconds). An example of its use may be for beer/ale fermentation. When the batch reaches a certain pH, the relay contacts would close and by some means (light, horn, etc.) alert someone to take a sample, or provide the trigger to automatically take a sample of the batch. The utility of this function can, of course, be expanded beyond sampling and be used whenever a timed relay output closure is required when the PV reaches a certain set point.

Interlock Relay(s)

This function allows a process to use one or more very low voltage input signals or simple switch contacts to control the state of one or more internal “interlock” relays. A violation (i.e. loss of input, open switch, or open circuit) forces one or more N/O interlock relay contacts to open. One input can be used in series with a number of interlock switches, or up to eight inputs can be required to force-on one (or more) internal interlock relays. Requires PDA1044 Digital I/O module or use of on-board digital input F4. Please see *PROVu Series Safety Interlock Feature* whitepaper on our website for more information.

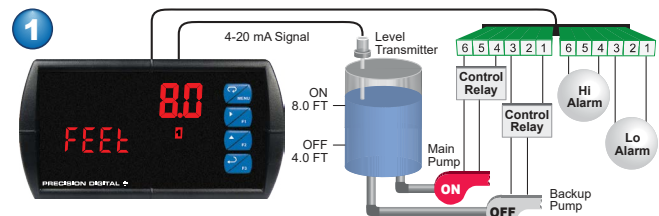


Switching Inductive Loads

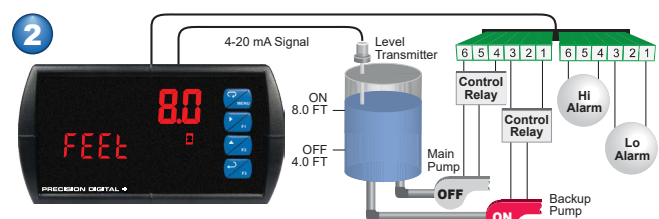
The use of suppressors (snubbers) is strongly recommended when switching inductive loads to prevent disrupting the microprocessor’s operation. The suppressors also prolong the life of the relay contacts. Precision Digital offers the PDX6901.

Multi-Pump Alternation

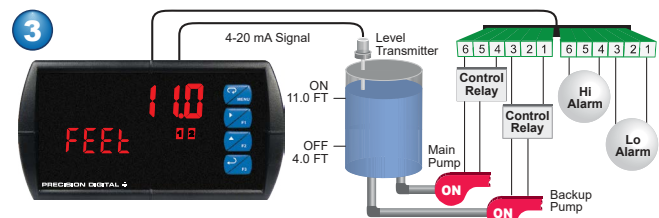
The PROVu can be used as a pump controller when combined with a continuous level transmitter. The most common pump control application is shown below: controlling and alternating two pumps and providing high and low-level alarms. The light/horn accessory can be added to provide visual and audible alarm indication.



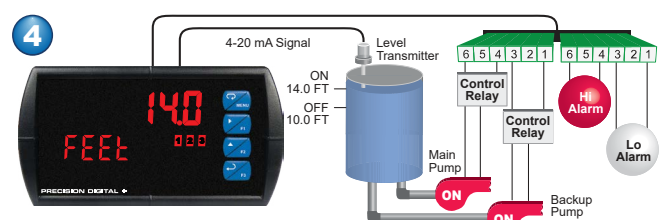
Relay #1 turns the main pump on at 8.0 feet and turns it off at 4.0 feet.



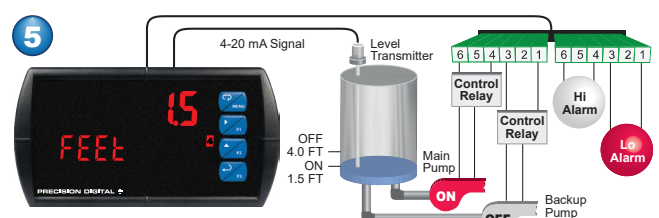
With the Pump Alternation feature activated, the next time the level reaches 8.0 feet, relay #2 transfers and starts the backup pump.



If the backup pump is not able to keep up, and the level reaches 11.0 feet, relay #1 transfers and starts the main pump as well.



Relay #3 trips the High-Level Alarm at 14.0 feet and resets at 10.0 feet.



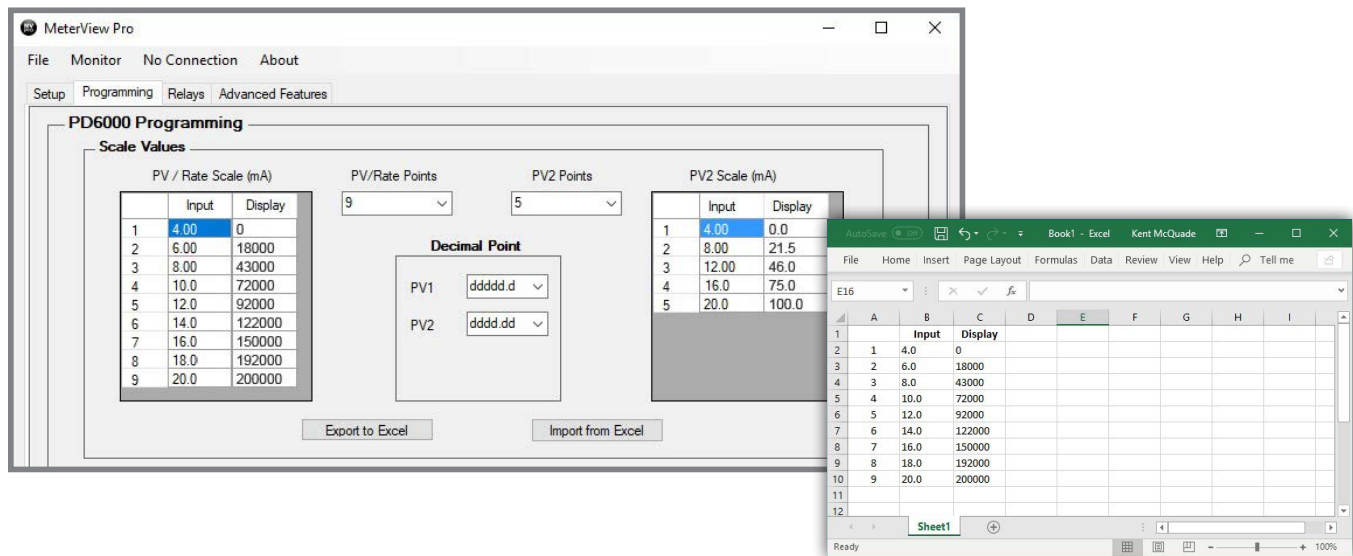
Relay #4 trips the Low-Level Alarm at 1.5 feet and resets at 4.0 feet.

SIGNAL INPUT CONDITIONING

There are many applications in the industrial world that can't be satisfied with simple, two-point linear scaling so the PROVu has advanced linearization capabilities to handle applications like round horizontal tank volume measurement, open channel flow, DP flow, and others. And all of these capabilities are easily programmed using MeterView Pro programming software.

32-Point Linearization

The most common way to linearize a non-linear signal is to break it up into smaller ranges that are more linear than the overall range. The PROVu is available with up to 32 points of linearization and if dual scale feature is used, the second PV can have up to eight points of linearization. The linearization data can be imported from an Excel spreadsheet or can be exported from MeterView Pro to an Excel spreadsheet. The following screen shot from MeterView Pro shows PV1 with 9 points of linearization and PV 2 with 5 points of linearization:



Scale values can also be imported from an Excel spreadsheet.

Specialized Linearization Functions

In addition to the generic 32- and 8-point linearization functions, the PROVu is also available with specialized functions for round horizontal tanks, open channel flow, and DP flow.

Round Horizontal Tank

Function: Signal Input Conditioning

RHT

Diameter: 48.000 Inch

Length: 120.000

The user enters the diameter and length of a flat-ended round horizontal tank resulting in a display of volume.

Programmable Exponent

Function: Signal Input Conditioning

Prog Exponent

Exponent: 1.683

The input is raised to an exponent programmable by the user resulting in a display of open channel flow rate.

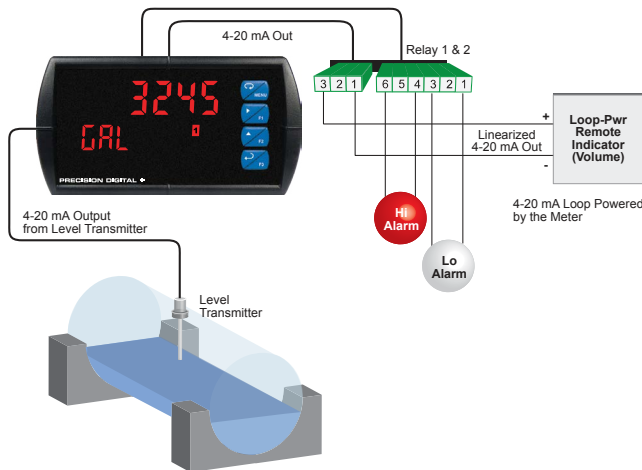
Square Root Extraction

Function: Signal Input Conditioning

Square Root

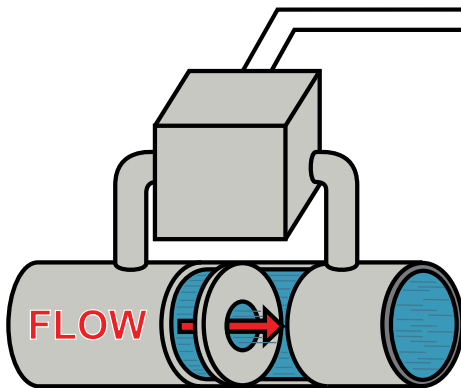
The square root of the input is taken resulting in a display of flow rate.

Round Horizontal Tank Volume Linearizer



In this application, a level sensor is measuring the height in the round horizontal tank and the PROVu is converting that signal to volume using the RHT function. All the user has to do is input the diameter and length of the tank and the meter converts the signal to volume.

DP Flow via Square Root Extraction

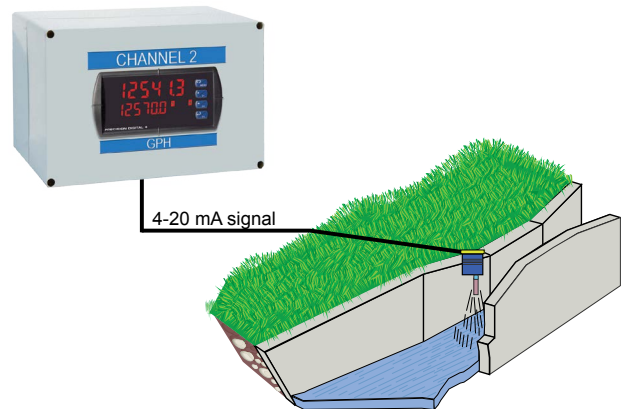


In this application, the PD6000 is displaying flow rate by extracting the square root from the 4-20 mA signal from a differential pressure transmitter. The user selectable low-flow cutoff feature gives a reading of zero when the flow rate drops below a user selectable value.

Linear 4-20 mA Analog Output

For applications where the input was linearized by the PROVu, the 4-20 mA output will represent that linearized value.

Open Channel Flow Rate Indication



In this application, a level sensor is measuring the height in a weir and the PROVu is converting that signal to flow rate using the programmable exponent function. All the user has to do is input the corresponding exponent for their weir and the meter will convert the signal to flow.

The following information is required for programming the PD6000 for open channel flow rate:

1. The exponent value associated with the flow calculation for the specific weir or flume being used.
2. The zero head, or water depth, mA value from the level transmitter.
3. The mA value from the transmitter for the maximum head, and the flow rate at that level. The level transmitter is normally programmed to provide 20 mA at the maximum head value and flow rate.

Example:

A 120" V-notch weir flow formula for millions of gallons per day is shown below.

$$MGD = 2.798 H^{2.5}$$

The exponent component is 2.5.

The level transmitter has been programmed so that at zero head, when the water level is at the base of the V-notch, the output is 4 mA.

The level transmitter has been programmed so that at the top of the V-notch, at 2.00 ft, the output is 20 mA.¹

The coefficient of the flow equation (2.798 in the above example), is not needed for programming the meter; the scaling function of the meter incorporates the coefficient and the head height automatically.

1. *Isco Open Channel Flow Measurement Handbook, Sixth Edition*, ed. Diane K. Walkowiak, M.A. (Teledyne Isco, Inc., 2006), 168-169.

PHYSICAL FEATURES

The PROVu is designed for ease-of-use in industrial applications. Considerations include a NEMA 4X front panel, wide operating temperature range, removable screw terminal connectors, snap in place mounting brackets, forgiving panel cutout requirement, and UL Listing for electrical safety. All of these features are backed by a 3-year warranty.

Type 4X / NEMA 4X Front Panel



Not only does the PROVu's front panel UL Type 4X approval indicate it is waterproof, but it also indicates it is rugged. Part of the UL Type 4X test is to drop a 2 inch solid stainless steel ball from 8 feet on top of the meter's faceplate.

Wide Operating Temperature Range

The PROVu can operate from -40 to 65°C (-40 to 150°F) meaning it can be installed in a wide variety of indoor and outdoor industrial applications. And over this range, the PROVu will drift no more than 0.005% of calibrated span/°C max from 0 to 65°C ambient and 0.01% of calibrated span/°C max from -40 to 0°C ambient.

Removable Screw Terminal Connectors

Industrial applications require screw terminal connections for easy field wiring and the PROVu goes one step further in convenience by making them removable also.



Easy Plug-in Removable Terminal Connectors



Secured-in-Place Rugged Mounting Brackets

If you're installing the PROVu outdoors in the hot or cold weather, the last thing you want to do is fumble around with mounting brackets that don't stay in place. The PROVu's mounting brackets can be easily secured into place and then screwed down to the panel. These brackets are rugged so they can be tightened to the panel to provide a solid NEMA 4X seal.



Easy Secured-in-Place Mounting Brackets

Forgiving Panel Cutout Requirement

The PROVu's bezel has been oversized to allow for not perfectly executed panel cutouts where NEMA 4X seal is not required.

Over-Sized Bezel to Completely Cover Panel Cutouts



UL Listing for Electrical Safety

UL & C-UL Listed: USA & Canada
UL 508 Industrial Control Equipment

UL File Number: E160849

Front Panel: UL Type 4X, NEMA 4X, IP65; panel gasket provided

Low Voltage Directive: EN 61010-1:2010 Safety requirements for measurement, control, and laboratory use

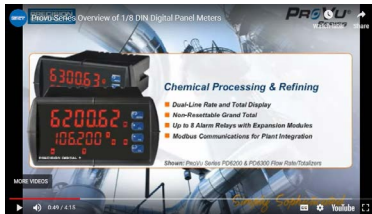
USB Port for Easy Connection to MeterView Pro Free Software



USB cable conveniently plugs into side of PROVu meter



VIDEOS TO WATCH



PROVu Series Overview

Learn About All the Meters in the PROVu Series!



PROVu Multi-Pump Alternation

Learn How to Use the PROVu as a Pump Controller.



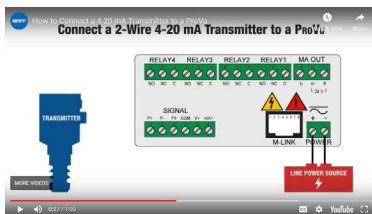
PROVu Function Keys

Learn How the PROVu's Function Keys Increase the Utility of the PROVu.



Connect a PROVu to a PC Using MeterView Pro

Learn How Easy it is to Use MeterView Pro Software.



Connect a 2-Wire 4-20 mA Transmitter to a PROVu

Learn How to Connect Your Transmitter to a PROVu.



Introduction to the Helios

Learn About the Large Display Version of the PROVu.

OPERATIONAL FEATURES

Function Keys, F4 Terminal, Digital Inputs

There are three ways the user can interact with the PROVu to perform a variety of useful functions:

1. Three Front Panel Function Keys

The default settings for the function keys are:



Reset Max/Min Reading



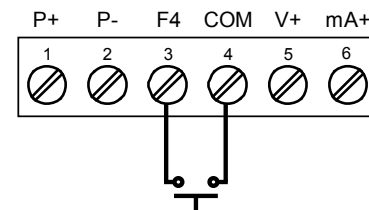
Display Max/Min Reading



Acknowledge Relays

2. F4 On-Board Digital Input

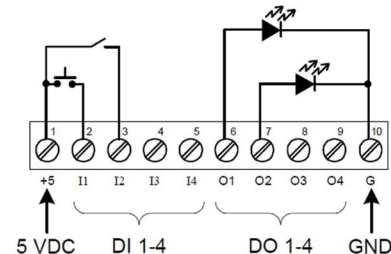
The PD6000 includes a digital input as standard. This digital input can operate with the tare, reset tare, or interlock relays feature, force relays on from a signal from a PLC or relay on other equipment, and much more. This is ideal for installations where the meter is inaccessible behind a cover, or where an additional function key is needed for customized operation.



The F4 terminal is particularly useful for wiring up a remote switch to reset the relays as shown here:



3. Optional 4 Digital Input/Output Module PDA1044



With these three methods, the PROVu can be programmed to trigger certain events (i.e acknowledge relays, reset max and/or min, disable/enable output relays, or hold current relay states), provide direct menu access points and more.

Function Key, Digital Inputs, & Digital Outputs Descriptions

The following table describes the actions that PROVu function keys and digital inputs can be programmed to perform. The table also describes how the digital outputs can be used to remotely monitor the PROVu's alarm relay states, or the states of a variety of actions and functions executed by the meter.

Display	Description	Item	Display	Description	Item
rSt H1	Reset the stored maximum display value	FK, DI, DO	Ln1 HL	Display maximum & minimum display values on line 1	FK, DI
rSt Lo	Reset the stored minimum display value	FK, DI, DO	Ln2 H1	Display maximum display value on line 2	FK, DI
rSt HL	Reset the stored maximum & minimum display values	FK, DI, DO	Ln2 Lo	Display minimum display value on line 2	FK, DI
tArE	Capture tare and zero the display	FK, DI, DO	Ln2 HL	Display maximum & minimum display values on line 2	FK, DI
rSt tr	Reset captured tare and resume normal operation	FK, DI, DO	F On 1*	Force relay 1 (*through 4) into the on state. This is used in conjunction with a digital input expansion module to achieve interlock functionality.	FK, DI
rELAY	Directly access the relay menu	FK, DI	ContrL	Directly access the control menu	FK, DI
SEt 1*	Directly access the set point menu for relay 1 (*through 8)	FK, DI	dISAbL	Disable the selected function key or digital I/O	FK, DI
rLY d	Disable all relays until a button assigned to enable relays (rLY E) is pressed	FK, DI	AcH	Acknowledge all active relays that are in a manual operation mode such as auto-manual or latching	FK, DI, DO
rLY E	Enable all relays to function as they have been programmed	FK, DI	rESEt	Directly access the reset menu	FK, DI
0 HoLd	Hold current relay states and analog output as they are until a button assigned to enable relays (rLY E) is pressed	FK, DI	mEnu	Mimic the menu button functionality (digital inputs only)	DI
d HoLd	Hold the current display value, relay states, and analog output momentarily while the function key or digital input is active. The process value will continue to be calculated in the background.	FK, DI	rIGHt	Mimic the right arrow/F1 button functionality (digital inputs only)	DI
Ln1 H1	Display maximum display value on line 1	FK, DI	uP	Mimic the up arrow/F2 button functionality (digital inputs only)	DI
Ln1 Lo	Display minimum display value on line 1	FK, DI	EntEr	Mimic the enter/F3 button functionality (digital inputs only)	DI
			ALAr 1*	Provide indication when alarm 1 (*through 8) has been triggered (digital outputs only)	DO

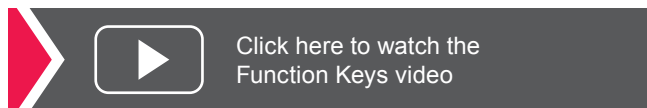
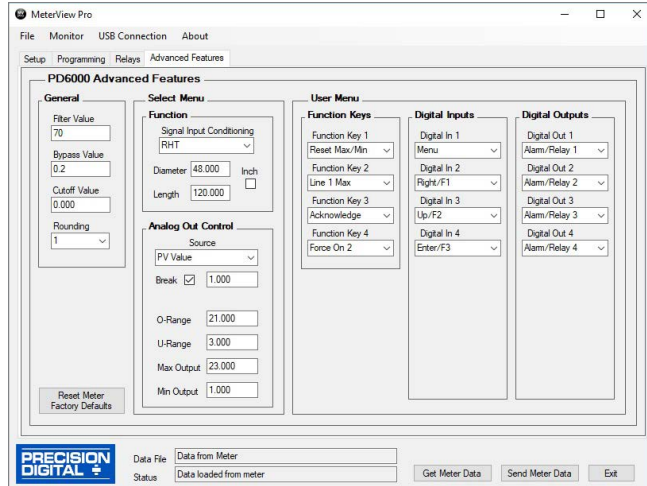
FK: Function Keys DI: Digital Inputs DO: Digital Outputs



Watch video about the programmable function keys, digital inputs, and all the capabilities these features offer on the PROVu Series.

Remote Operation of Front Panel Buttons

The user can operate the front panel buttons from a remote location by using digital inputs programmed in the following manner:



Tare

The tare function zero's out the display. In the case of scale weight, tare is used to eliminate container weight and provide net weight readings. There are two tare functions; Capture Tare and Reset Tare. When the capture tare function is used, the display reading is offset by the displayed amount to make the displayed value zero. This modified display value is the net value. The originally displayed value without the tare offset is the gross value. Both may be chosen as a display option. Reset tare removes the display offset.



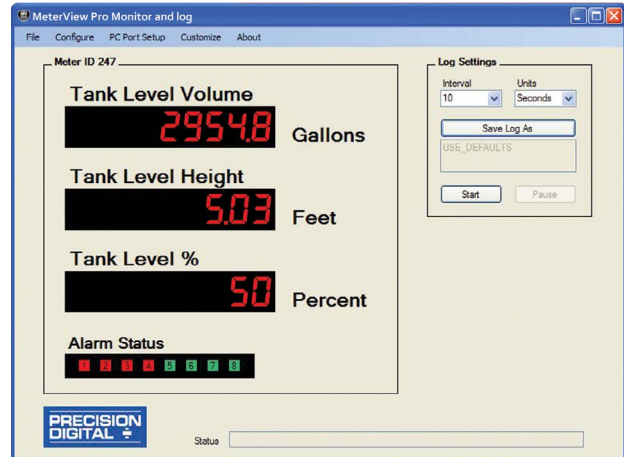
Reset tare removes the display offset of the net value, and the gross and net values become the same until a new capture tare is entered.

Max / Min Display

Max/Min (or Peak/Valley) is standard on the ProVu PD6000. Either display can be configured to show either maximum or minimum excursion since last reset. The displays can also be configured to toggle between Max and Min values. Both values can be simply reset from the front panel.

MeterView Pro Monitoring & Datalogging Software

Not only does free MeterView Pro software greatly simplify setup and programming of the ProVu, it can also be used to monitor and datalog your process.



- Custom Tags: i.e. Tank Level Volume
- Custom Units: i.e. Gallons, Feet, Percent
- Alarm Status Indicators

Datalog Report

Collected data logger information can be sent to a CSV file for importing into a spreadsheet program. Below is an example of one such file. Of course, once within the spreadsheet, much can be done to customize the data.

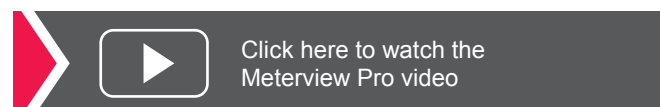
w ProFurnace 3.csv Created 8/10/2010 3:22:37 PM

COM5Logging Rate:1 update every 5Seconds

Date & Time	Tag1	Display	Units	Tag2	Display	Units	Tag3	Display	Units	R1	R2	R3	R4
8/10/2010 3:22	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:22	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:22	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:22	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:22	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:22	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:23	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:24	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off
8/10/2010 3:24	Furn 3	207	Degrees F	MAX	247	Degrees F	MIN	206	Degrees F	Off On	Off Off	Off Off	Off Off

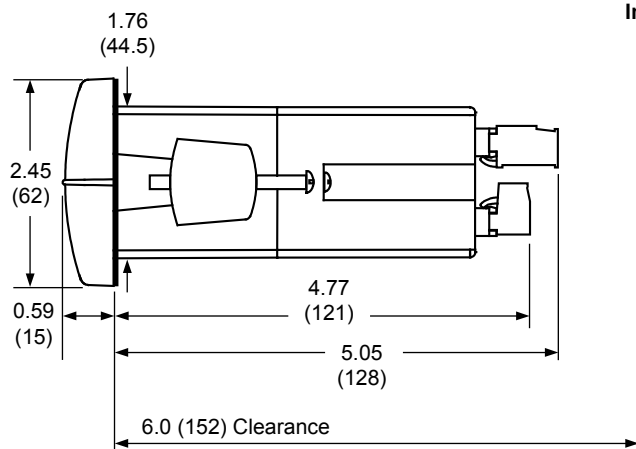
Relay Control

Relays can be controlled from MeterView Pro for testing purposes. This is commonly done to determine whether the relays are functioning properly. In the *Setup* window, under *Relay and Digital Out Test* you have the option of selecting the relays you want in an ON state or OFF state and also whether you want to leave the relays in manual control or to return them to automatic operation.

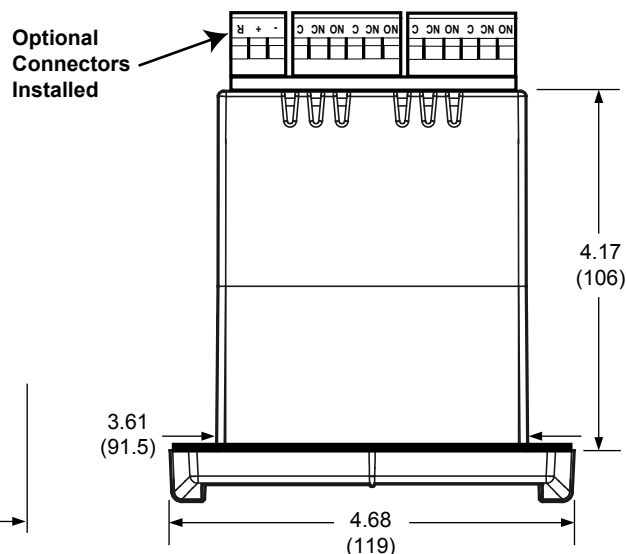


DIMENSIONS

Units: Inches (mm)



Side View



Top View

Notes:

1. Panel cutout required: 1.772" x 3.622" (45 mm x 92 mm)
2. Panel thickness: 0.040 - 0.250" (1.0 mm - 6.4 mm)
3. Mounting brackets lock in place for easy mounting
4. Clearance: Allow 6" (152 mm) behind the panel

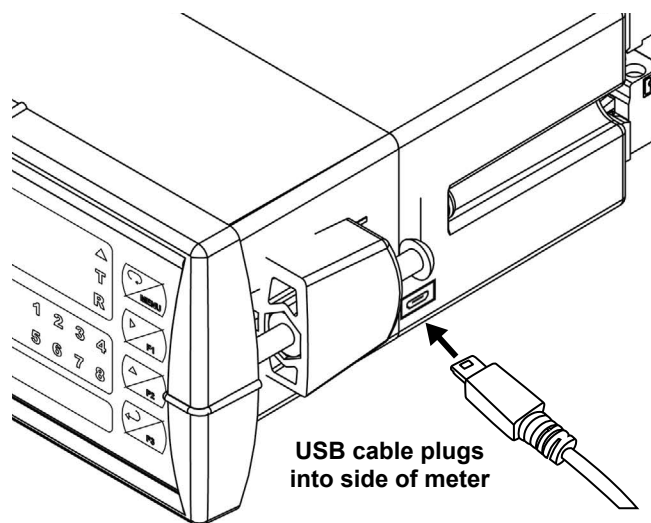
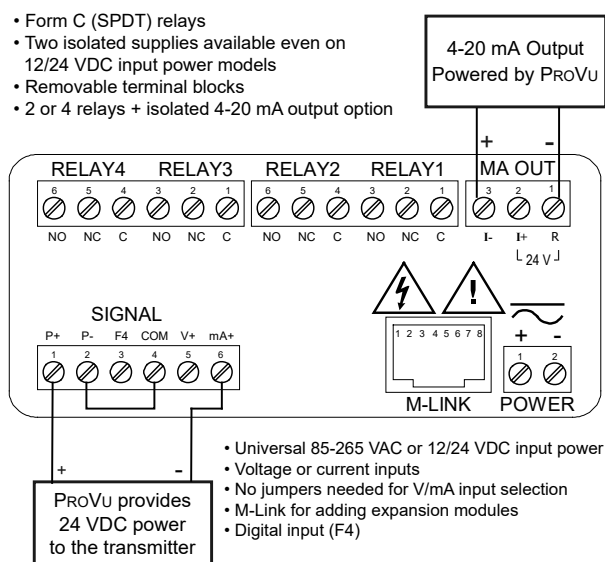


Download free 3-D CAD files of these instruments to simplify your drawings!

predig.com/documentation-cad

CONNECTIONS

- Form C (SPDT) relays
- Two isolated supplies available even on 12/24 VDC input power models
- Removable terminal blocks
- 2 or 4 relays + isolated 4-20 mA output option



USB cable plugs into side of meter



Click here to watch video on how to connect a 2-wire transmitter to a ProVu

SPECIFICATIONS

Except where noted all specifications apply to operation at +25°C.

General

Display	Line 1: 0.60" (15 mm) high, red LEDs Line 2: 0.46" (12 mm) high, red LEDs 6 digits each (-99999 to 99999), with lead zero blanking
Display Intensity	Eight user selectable intensity levels. Default value is six.
Display Update Rate	5/second (200 ms)
Overrange	Display flashes 99999
Underrange	Display flashes -99999
Display Assignment	Display Line 1: PV1, PV2, PCT, PV & units, gross weight, net & gross weight, max/min, max & min, set points, or Modbus input. Display Line 2: Same as Display Line 1; plus units, tag or turned off.
Programming Methods	Four front panel buttons, digital inputs, PC and MeterView Pro software, or Modbus registers.
Noise Filter	Programmable from 2 to 199 (0 will disable filter)
Filter Bypass	Programmable from 0.1 to 99.9% of calibrated span
Recalibration	All ranges are calibrated at the factory. Recalibration is recommended at least every 12 months.
Max/Min Display	Max/min readings reached by the process are stored until reset by the user or until power to the meter is turned off.
Rounding	Select 1, 2, 5, 10, 20, 50, or 100 (e.g. rounding = 10, value = 123.45, display = 123.50).
Tare	Tare function zeros out the meter to remove the for weight of a container. Tare function can be assigned to a function key, F4 terminal, or a digital input.
Password	Three programmable passwords restrict modification of programmed settings.
Non-Volatile Memory	All programmed settings are stored in non-volatile memory for a minimum of ten years if power is lost.
Power Options	85-265 VAC 50/60 Hz; 90-265 VDC, 20 W max; 12-24 VDC, 12-24 VAC, 15 W max. Powered over USB for configuration only.
Fuse	Required external fuse: UL Recognized, 5 A max, slow blow; up to 6 meters may share one 5 A fuse
Normal Mode Rejection	Greater than 60 dB at 50/60 Hz
Isolation	4 kV input/output-to-power line 500 V input-to-output or output-to-P+ supply
Overvoltage Category	Installation Overvoltage Category II: Local level with smaller transient overvoltages than Installation Overvoltage Category III.

Environmental	Operating temperature range: -40 to 65°C (-40 to 149°F) Storage temperature range: -40 to 85°C (-40 to 185°F) Relative humidity: 0 to 90% non-condensing
Connections	Removable screw terminal blocks accept 12 to 22 AWG wire, RJ45 for external relays, digital I/O, and serial communication adapters.
Enclosure	1/8 DIN, high impact plastic, UL 94V-0, color: black
Front Panel	NEMA 4X, IP65
Mounting	1/8 DIN panel cutout required: 3.622" x 1.772" (92 mm x 45 mm) Two panel mounting bracket assemblies are provided.
Tightening Torque	Screw terminal connectors: 5 lb-in (0.56 Nm)
Overall Dimensions	4.68" x 2.45" x 5.64" (119 mm x 62 mm x 143 mm) (W x H x D)
Weight	9.5 oz (269 g)
Warranty	3 years parts & labor. See Warranty Information and Terms & Conditions on www.predig.com for complete details.

Process Input

Inputs	Field selectable: 0-20 mA, 4-20 mA ±10 V (0-5 V, 1-5 V, 0-10 V) Modbus PV (Slave)
Isolated Transmitter Power Supply	Terminals P+ & P-: 24 VDC ±10%. All models selectable for 24, 10, or 5 VDC supply (internal jumper J4). 85-265 VAC models rated @ 200 mA max, 12-24 VDC powered models rated @ 100 mA max. 5 & 10 VDC supply rated @ 50 mA max. Refer to Transmitter Supply Voltage Selection (P+, P-) on page 17. When the Light/Horn is powered by the transmitter power supply, see MOD-LH Light/ Horn's transmitter power supply specification on page 28 for additional details. Light/Horn power not available for 5 or 10 VDC supplies.
Accuracy	±0.03% of calibrated span ±1 count, square root & programmable exponent accuracy range: 10-100% of calibrated span
Temperature Drift	0.005% of calibrated span/°C max from 0 to 65°C ambient, 0.01% of calibrated span/°C max from -40 to 0°C ambient
Input Signal Conditioning	Linear, square root, programmable exponent, or round horizontal tank volume calculation
Multi-Point Linearization	2 to 32 points for PV or PV1 2 to 8 points for PV2 (Dual-scale Level feature)
Programmable Exponent	User selectable from 1.0001 to 2.9999 for open channel flow

Round Horizontal Tank	Diameter & Length: 999.999 inch or cm calculates volume in gallons or liters respectively.	
Low-Flow Cutoff	0.1 to 999,999 (0 disables cutoff function). Point below at which display always shows zero.	
Decimal Point	Up to five decimal places or none: <i>dddddd, dddddd, dddd, ddd, dd, or dddddd</i>	
Calibration Range	Input Range	Minimum Span Input 1 & 2
	4-20 mA	0.15 mA
	±10 V	0.10 V
	An error message will appear if the input 1 and input 2 signals are too close together.	
Input Impedance	Voltage ranges: greater than 500 kΩ Current ranges: 50 - 100 Ω (depending on internal resettable fuse impedance)	
Input Overload	Current input protected by an internal resettable fuse, 30 VDC max. Fuse resets automatically after fault is removed.	
HART Transparency	The meter does not interfere with existing HART communications; it displays the 4-20 mA primary variable and it allows the HART communications to pass through without interruption. The meter is not affected if a HART communicator is connected to the loop. The meter does not display secondary HART variables.	

Relays

Rating	2 or 4 SPDT (Form C) internal and/or 4 SPDT (Form A) external; rated 3 A @ 30 VDC and 125/250 VAC resistive load; 1/14 HP (≈ 50 W) @ 125/250 VAC for inductive loads	
Noise Suppression	Noise suppression is recommended for each relay contact switching inductive loads.	
Deadband	0-100% of span, user programmable	
High or Low Alarm	User may program any alarm for high or low trip point. Unused alarm LEDs and relays may be disabled (turn off).	
Relay Operation	Automatic (non-latching), latching (requires manual acknowledge) with/without clear, sampling (based on time), pump alternation control (2 to 8 relays), Off (disable unused relays and enable interlock feature, manual on/off control mode).	
Relay Reset (Acknowledge)	User selectable via front panel buttons or digital inputs. - Automatic reset only (non-latching), when input passes the reset point. - Automatic + manual reset at any time (non-latching). - Manual reset only, at any time (latching). - Manual reset only after alarm condition has cleared (latching). <i>Note: Front panel button or digital input may be assigned to acknowledge relays programmed for manual reset.</i>	

Time Delay	0 to 999.9 seconds, on & off relay time delays. Programmable and independent for each relay
Fail-Safe Operation	Programmable and independent for each relay. <i>Note: Relay coil is energized in non-alarm condition. In case of power failure, relay will go to alarm state.</i>
Auto Initialization	When power is applied to the meter, relays will reflect the state of the input to the meter
Additional Relays	An external module, model PDA1004 , is available to add 4 SPST 3 A relays to the meter.

Isolated 4-20 mA Transmitter Output

Output Source	Process variable (PV), max, min, set points 1-8, Modbus input, or manual control mode		
Scaling Range	1.000 to 23.000 mA for any display range		
Calibration	Factory calibrated: 4.000 to 20.000 = 4-20 mA output		
Analog Out Programming	23.000 mA maximum for all parameters: Overrange, underrange, max, min, and break		
Accuracy	± 0.1% of span ± 0.004 mA		
Temperature Drift	0.4 µA/°C max from 0 to 65°C ambient, 0.8 µA/°C max from -40 to 0°C ambient <i>Note: Analog output drift is separate from input drift.</i>		
Isolated Transmitter Power Supply	Terminals I+ & R: 24 VDC ±10%. May be used to power the 4-20 mA output or other devices. All models rated @ 40 mA max.		
External Loop Power Supply	35 VDC maximum		
Output Loop Resistance	Power supply	Minimum	Maximum
	24 VDC	10 Ω	700 Ω
	35 VDC (external)	100 Ω	1200 Ω
Additional 4-20 mA Outputs	The PD659-1MA-2MA can split the optional 4-20 mA output into two isolated 4-20 mA outputs		
0-10 VDC Output	The PD659-1MA-1V can convert the optional 4-20 mA output to a 0-10 VDC output		

USB Connection

Function	Programming only
Compatibility	USB 2.0 Standard, Compliant
Connector Type	Micro-B receptacle
Cable	USB A Male to Micro-B Cable
Driver	Microsoft® Windows® 7/1/110
Power	USB port provides power to the meter. DO NOT apply AC or DC power to the meter while the USB port is in use.

On-Board Digital Input (F4)

Function	Remote operation of front-panel buttons, acknowledge/reset relays, reset max/min values.
Contacts	3.3 VDC on contact. Connect normally open contacts across F4 to COM.
Logic Levels	Logic High: 3 to 5 VDC Logic Low: 0 to 1.25 VDC
Additional I/O	Up to 2 external modules, model PDA1044 with 4 digital inputs and 4 digital outputs each can be added.

Modbus RTU Serial Communications

Slave Id	1 – 247 (Meter address)
Baud Rate	300 – 19,200 bps
Transmit Time Delay	Programmable between 0 and 199 ms
Data	8 bit (1 start bit, 1 or 2 stop bits)
Parity	Even, Odd, or None with 1 or 2 stop bits
Byte-To-Byte Timeout	0.01 – 2.54 second
Turn Around Delay	Less than 2 ms (fixed)

Note: Refer to the ProVu Modbus Register Tables located at www.predig.com for details.

MeterView Pro Software

Availability	Download directly from meter or from www.predig.com/download_software
System Requirements	Microsoft® Windows® 7/10/11
Communications	USB 2.0 (for programming only) (Standard USB A to Micro USB B) RS-232 adapter, RS-485 adapter and RS-485 to USB converter (programming, monitoring, and data logging)
Configuration	Configure meters one at a time
Power	USB port provides power to the meter. DO NOT apply AC or DC power to the meter while the USB port is in use.

Digital I/O Expansion Module

Channels	4 digital inputs & 4 digital outputs per module
System	Up to 2 modules for a total of 8 inputs & 8 outputs
Digital Input Logic	High: 3 to 5 VDC Low: 0 to 1.25 VDC
Digital Output Logic	High: 3.1 to 3.3 VDC Low: 0 to 0.4 VDC
Source Current	10 mA maximum
Sink Current	1.5 mA minimum
+5 V Terminal	To be used as pull-up for digital inputs only.

4-Relay Expansion Module

Relays	Four Form A (SPST) rated 3 A @ 30 VDC and 125/250 VAC resistive load; 1/14 HP (≈ 50 watts) @ 125/250 VAC for inductive loads.
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MOD-LH Light/Horn

Light Colors	MOD-LHRB1: Red MOD-LHGB1: Green MOD-LHYB1: Yellow MOD-LHBB1: Blue MOD-LHWB1: White MOD-LH5CB1: User selectable: red, green, yellow, blue, white MOD-LH3CB1-RYG: 1 layer each of red, yellow, green (consult factory for other colors available)
Light Action	Can be wired to flash (not available on MOD-LH5CB1) or stay steady on
Light/Horn & Button Installation	When MOD-LH is ordered with an enclosure, the user performs installation and wiring of Light/Horn and Button in pre-drilled holes.
Horn	85 dB
Rating	IP 65
Light/Horn Independence	Light and horn can be controlled via separate relays
Power Requirement	No additional power required when wired to a ProVu meter. When mounted remote: 24 VDC
Transmitter Power Supply	The ProVu's internal transmitter power supply is capable of supplying 200 mA to power the transmitter and other devices such as the Light/Horn. The following table illustrates how much of this power is required to drive various Light/Horns. If more power is needed, then consider the PDA1024-01.

MOD-LH and MOD-LH5CB1 Models:

Color	Power Required	Color	Power Required
Red	17 mA	Blue	15 mA
Green	15 mA	White	42 mA
Yellow	23 mA	Horn	20 mA

Example: 17 mA (Red Light) + 20 mA (Horn) = 37 mA total current needed from the 200 mA supply. Available current = 163 mA

MOD-LH3LCB1-RYG:

Power Requirement for the horn and each color that is turned on:

Color	Power Required	Color	Power Required
Red	34 mA	Yellow	33 mA
Green	29 mA	Horn	38 mA

Example: 33 mA (Yellow Light) + 38 mA (Horn) = 71 mA total current needed from the 200 mA supply. Available current = 139 mA

Reset / Silence Button	NEMA 4X; may be wired to F4 terminal on ProVu. F3 front panel button can also be used to reset relays.
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Button Labels	The Light/Horn accessory comes with 9 pre-printed message labels the user can affix under the red button: RESET, BATCH, ACK, TARE, SILENCE, STOP, START, PAUSE, START/STOP
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Light/Horn Mounting Connection	M22
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Hole Sizes	Light/Horn: 0.875" (22 mm) Button: 0.630" (16 mm)
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Cable Length:	3.28 feet (1 meter)
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Operating Temperature Range	-5 to 40°C (23 to 104°F)
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Compliance Information

Safety

UL & C-UL Listed	USA & Canada UL 508 Industrial Control Equipment
UL File Number	E160849
Front Panel	UL Type 4X, NEMA 4X, IP65; panel gasket provided
Low Voltage Directive	EN 61010-1 Safety requirements for measurement, control, and laboratory use

Electromagnetic Compatibility

Emissions	EN 55022 Class A ITE emissions requirements
Radiated Emissions	Class A
AC Mains Conducted Emissions	Class A
Immunity	EN 61326-1 Measurement, control, and laboratory equipment EN 61000-6-2 EMC heavy industrial generic immunity standard
RFI - Amplitude Modulated	80 - 1000 MHz 10 V/m 80% AM (1 kHz) 1.4 - 2.0 GHz 3 V/m 80% AM (1 kHz) 2.0 - 2.7 GHz 1 V/m 80% AM (1 kHz)
Electrical Fast Transients	±2kV AC mains, ±1kV other
Electrostatic Discharge	±4kV contact, ±8kV air
RFI - Conducted	10V, 0.15-80 MHz, 1kHz 80% AM
AC Surge	±2kV Common, ±1kV Differential
Surge	1KV (CM)
Power-Frequency Magnetic Field	30 A/m 70%V for 0.5 period
Voltage Dips	40%V for 5 & 50 periods 70%V for 25 periods
Voltage Interruptions	<5%V for 250 periods

Note: Testing was conducted on meters installed through the covers of grounded metal enclosures with cable shields grounded at the point of entry representing installations designed to optimize EMC performance.

EU Declaration of Conformity

For shipments to the EU and UK, a Declaration of Conformity was printed and included with the product. For reference, a Declaration of Conformity is also available on our website www.predig.com/docs.

PD6000 ProVu Dual-Line 6-Digit Process Meter

ORDERING INFORMATION

ProVu PD6000 • Standard Models		
85-265 VAC Model	12-24 VDC Model	Options Installed
PD6000-6R0	PD6000-7R0	None
PD6000-6R2	PD6000-7R2	2 Relays
PD6000-6R3	PD6000-7R3	4-20 mA Output
PD6000-6R4	PD6000-7R4	4 Relays
PD6000-6R5	PD6000-7R5	2 Relays & 4-20 mA Output
PD6000-6R7	PD6000-7R7	4 Relays & 4-20 mA Output

Note: 24 V Transmitter power supply standard on all models.

ProVu PD6000 • SunBright Display Models		
85-265 VAC Model	12-24 VDC Model	Options Installed
PD6000-6H0	PD6000-7H0	None
PD6000-6H2	PD6000-7H2	2 Relays
PD6000-6H3	PD6000-7H3	4-20 mA Output
PD6000-6H4	PD6000-7H4	4 Relays
PD6000-6H5	PD6000-7H5	2 Relays & 4-20 mA Output
PD6000-6H7	PD6000-7H7	4 Relays & 4-20 mA Output

Note: 24 V Transmitter power supply standard on all models.

Accessories	
Model	Description
MOD-LHRB1	Red ⁽²⁾ Light/Horn and Button with Holes Drilled for Light/Horn and Button in Enclosure ⁽¹⁾
PDA-BUTTON1R	Button
PDA-LHR	Red ⁽²⁾ Light/Horn
PDA1002	DIN Rail Mounting Kit for Two Expansion Modules
PDA1004	4-Relay Expansion Module
PDA1024-01	24 VDC Power Supply for DIN Rail
PDA1044	4 Digital Inputs & 4 Digital Outputs Module
PDA1232	RS-232 Serial Adapter
PDA1485	RS-485 Serial Adapter
PDA18DINSH	Stainless Steel Sun Hood
PDA7485-I	RS-232 to RS-422/485 Isolated Converter
PDA8232-N	USB to RS-232 Non-Isolated Converter
PDA8485-I	USB to RS-422/485 Isolated Converter
PDX6901	Suppressor (snubber): 0.01 μ F/470 Ω , 250 VAC

1. The enclosure comes pre-drilled with holes for Light/Horn and Button to be installed by user. Meter / controller and enclosure are sold separately. The Light/Horn hole is located on the top in the back left corner of the enclosure and the button hole is centered on the front of the enclosure about an inch off the bottom of the door. For mounting in different locations, order items separately and drill holes and mount as desired.
2. For other light color options see the MOD-LH Series manual (LIMMODLH).

ProVu Upgrade Cards	
Model	Description
PD1102	2 Relays
PD1103	4-20 mA Output ¹
PD1104	4 Relays
PD1105	2 Relays + 4-20 mA Output ¹
PD1107	4 Relays + 4-20 mA Output ¹

1. Output calibration required by user.
2. These upgrade cards are intended for customers who already have a meter and want to upgrade its functionality.

PDA2360 Series Control Stations	
Model	Description
PDA2360-E	Emergency Stop Button
PDA2361-A	1 Black Ack Button
PDA2361-Q	1 Black Silence Button
PDA2364-MRUE	4 Black Buttons: Menu, Right, Up, Enter

Your Local Distributor is:

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